
Dynamic Principal Intervention in Static Mean-Field Games: From Annealed TRPA to Bilevel Stackelberg Learning

Jin Ying

Zhao Yutai

Abstract

We study independent learning in static mean-field games (SMFG) through the lens of variational inequalities. Building on the **TRPA-Full** (Tikhonov-Regularized Projected Ascent, full feedback) algorithm of Yardim et al. [2025], which attains an $\mathcal{O}(N^{-1/4})$ exploitability floor under constant Tikhonov regularization, we introduce **Annealed-TRPA (A-TRPA)**: the single algorithmic modification of using the annealed Tikhonov schedule $\tau_t = \eta_t = t^{-1/2}$. A-TRPA outputs the weighted averaged iterate $\bar{\pi}_T^i = (\sum_t \eta_t \pi_t^i) / (\sum_t \eta_t)$. We prove that the deterministic population-and-expectation average $\hat{\pi}_T = (1/N) \sum_i \mathbb{E}[\bar{\pi}_T^i]$ is, under homogeneous deployment, an $\tilde{\mathcal{O}}(T^{-1/2} + N^{-1/2} + \sigma)$ -Nash equilibrium of the N -player game; under noiseless feedback ($\sigma = 0$) this becomes $\tilde{\mathcal{O}}(T^{-1/2} + N^{-1/2})$, matching the deployment-level floor of Yardim et al. [2025, Theorem 1] and closing the $N^{1/4}$ gap left open by Yardim’s Corollary 1. The last iterate also tracks the same rate empirically; a rigorous last-iterate bound at the annealed schedule is open due to a specific technical obstacle (Appendix B). As an algorithmic extension, we present **B-ATRPA** for the bilevel **Stackelberg-MFG** setting, embedding A-TRPA inside a symmetric two-point Flaxman–Kalai–McMahan zeroth-order ascent at the principal level. B-ATRPA is evaluated empirically; a sufficient-condition sketch toward a non-asymptotic outer-loop guarantee is given in the appendix. Empirical verification on Linear, KL, and Beach-Bar benchmarks shows that A-TRPA-Avg’s N -scaling slope is -0.48 on Linear and -0.46 on Beach-Bar (theoretical -0.5); TRPA-Full’s empirical curve sits above A-TRPA-Avg on Linear and Beach-Bar across all N (KL is a draw); and B-ATRPA’s welfare improves from -0.17 to -0.05 versus a flat no-principal baseline at -0.17 .

1 Introduction

1.1 Background: Mean-Field Games and Static Mean-Field Games

Strategic interaction in a population of N agents is generally intractable: as N grows, the joint Nash equilibrium of the N -player game becomes computationally and analytically out of reach. Mean-field games (MFG) [Lasry and Lions, 2007, Huang et al., 2006] address this curse of dimensionality by replacing the finite population with a continuum limit, in which a representative agent optimizes against the aggregate population distribution rather than against every other player individually.

Classical MFGs model temporal state evolution through coupled Hamilton–Jacobi–Bellman and Fokker–Planck equations. In many large-population systems, however, the dominant interaction can be approximated by the current aggregate load or allocation, as in congestion, routing, bandwidth sharing, or load balancing. This stateless regime is the setting of static mean-field games (SMFGs), closely related to nonatomic congestion and routing games [Rosenthal, 1973, Beckmann et al., 1956], population games and evolutionary dynamics [Sandholm, 2010], and mean-field bandits [Gummadi et al., 2013].

Beyond statelessness, the setting we study is decentralized. Agents do not coordinate their updates, communicate with one another, or observe the full population strategy. Instead, they learn from payoff feedback generated by finite-population play. This is the setting studied by Yardim et al. [2025], whose variational inequality (VI) formulation is the direct starting point for our analysis.

We adopt this VI framework because the gap studied in this paper is a regularization effect in projected VI learning. In Yardim et al. [2025], Tikhonov-Regularized Projected Ascent with full feedback (TRPA-Full) is a decentralized projected-learning algorithm: each agent observes its own noisy payoff vector over all actions and updates by a Tikhonov-regularized projection step. The Tikhonov term stabilizes finite-population feedback noise, but its constant strength also biases the learned solution away from the original SMFG equilibrium condition.

This bias appears explicitly in the finite-population guarantees of Yardim et al. [2025]. Their Theorem 1 shows that homogeneous deployment of an exact mean-field equilibrium π^* achieves an $\mathcal{O}(1/\sqrt{N})$ -NE of the N -player game, providing the finite-population deployment benchmark. By contrast, their Corollary 1 shows that TRPA-Full with the optimal constant Tikhonov parameter $\tau = N^{-1/4}$ attains only $\mathcal{O}(N^{-1/4})$ exploitability. The resulting factor- $N^{1/4}$ gap motivates our main question: is this loss a fundamental price of decentralized learning, or an artifact of using a constant regularization schedule?

We argue that the gap is an artifact of the constant Tikhonov parameter. The optimal constant choice $\tau = N^{-1/4}$ balances regularization bias $\mathcal{O}(\tau)$ against finite-population variance $\mathcal{O}(\tau^{-1}/\sqrt{N})$, leaving both terms at order $N^{-1/4}$. Replacing the constant parameter by an annealed schedule $\tau_t \rightarrow 0$ removes the persistent bias, but only if the schedule decays slowly enough to keep the variance controlled. We answer this question by analyzing an annealed version of TRPA-Full.

Finally, the annealed Tikhonov signal also admits an intervention interpretation: it acts as a time-varying population-level incentive that gradually vanishes. This observation motivates our bilevel Stackelberg extension (Section 4), where a principal optimizes such interventions from zeroth-order welfare feedback while the population continues to learn independently.

1.2 Our Contributions

We introduce **A-TRPA**, a variant of TRPA-Full in which the constant τ is replaced by the time-decaying schedule $\tau_t = \eta_t = t^{-1/2}$. Our contributions are:

1. **Interpretable insight.** We provide a *principal-intervention* interpretation of the Tikhonov term in TRPA: the update direction $r_t^i - \tau_t \pi_t^i$ decomposes into a payoff-driven gradient r_t^i and a centralized ℓ^2 pull $-\tau_t \pi_t^i$ toward the uniform distribution. A time-varying τ_t thus realizes a *dynamic* principal intervention modulating exploration–exploitation over time without communication.
2. **Tighter convergence guarantee.** A-TRPA outputs the weighted averaged iterate $\bar{\pi}_T^i = (\sum_t \eta_t \pi_t^i) / (\sum_t \eta_t)$. Under the annealed Tikhonov schedule $\tau_t = \eta_t = t^{-1/2}$, the deterministic population-and-expectation average $\hat{\pi}_T = (1/N) \sum_i \mathbb{E}[\bar{\pi}_T^i]$ is, under homogeneous deployment, an $\tilde{\mathcal{O}}(T^{-1/2} + N^{-1/2} + \sigma)$ -Nash equilibrium, closing the $N^{1/4}$ gap of TRPA-Full in the noiseless ($\sigma = 0$) case and matching the deployment-level $N^{-1/2}$ floor of Theorem 1 of Yardim et al. [2025]. We additionally track the realized last iterate as an empirical diagnostic and observe matching scaling; a rigorous last-iterate bound is open and the technical obstacle is identified in Appendix B.
3. **Bilevel algorithmic extension.** We propose **B-ATRPA** for a *bilevel Stackelberg-MFG* in which a principal optimizes a welfare functional over its own action $w \in W$ while the population learns via A-TRPA. The principal runs a symmetric two-point sphere-smoothed zeroth-order ascent on noisy welfare evaluations. B-ATRPA is presented as an algorithmic extension and evaluated empirically; a sufficient-condition sketch identifying when standard non-convex zeroth-order analysis yields a $\tilde{\mathcal{O}}(\epsilon^{-12}N)$ budget toward ϵ -stationarity of the smoothed welfare is provided in Appendix C. A complete non-asymptotic outer-loop convergence proof is left for future work.
4. **Empirical validation.** On the LINEAR, KL, and Beach-Bar benchmarks of Yardim et al. [2025], A-TRPA empirically matches the predicted $\mathcal{O}(N^{-1/2})$ scaling (A-TRPA-Avg Linear slope -0.48 , BB slope -0.46 ; A-TRPA-Last is similar; 10 seeds) and outperforms TRPA on Linear and BB across all population sizes $N \in \{20, 50, 100, 200, 500, 1000\}$ and at both noise levels $\sigma \in \{0, 0.1\}$ (KL last-iterate is comparable, with TRPA marginally lower at $N = 1000$ within seed noise). B-ATRPA’s welfare convergence on Stackelberg-Linear improves the welfare from -0.16 to -0.05 , while a same-budget random-principal baseline produces no measurable improvement (final welfare -0.17 , identical to the no-principal baseline), confirming that the gain comes from the FKM gradient direction rather than from perturbing w .

1.3 Related Work

Variational inequalities. Our analysis builds on the VI literature for monotone operators [Facchinei and Pang, 2003]. Standard first-order methods include extragradient [Korpelevich, 1976], proximal point [Rockafellar, 1976], and mirror-prox / prox-method [Nemirovski, 2004]; a separate line studies iterative Tikhonov regularization for stabilizing projected-gradient-type methods, with either a constant or a prescribed decay schedule for the regularizer [Kotsalis et al., 2022, Beznosikov et al., 2023]. Our time-decaying schedule is closer in spirit to annealing in stochastic approximation [Robbins and Monro, 1951] and to exploration schedules in online learning [Cesa-Bianchi and Lugosi, 2006]. We emphasize, however, that classical VI methods typically assume a single algorithmic process with (possibly noisy) oracle access to the operator $\mathbf{F}$. How N mutually non-communicating agents can collectively decay a shared regularization parameter τ_t while maintaining convergence is a question without a direct precedent in this literature, and is the central technical challenge we address.

Learning in mean-field games. A first line of MFG-learning work alternates between policy optimization and population-distribution updates: finite-state fictitious play [Perrin et al., 2020], online mirror descent under monotonicity [Pérolat et al., 2022], potential-MFG methods [Geist et al., 2022], and two-timescale Q-learning [Angiuli et al., 2021, 2023]. These approaches typically rely on monotonicity, potential structure, or contractivity, together with repeated access to or estimation of the mean-field distribution. Yardim et al. [2025] instead analyze independent learning directly in the static MFG through the VI formulation, providing, to our knowledge, the first finite-sample bounds under both full and bandit feedback in the no-communication setting. While their result is a fundamental advance, the constant Tikhonov regularization that makes their analysis tractable also pins the convergence rate at $\mathcal{O}(N^{-1/4})$: for any fixed N , this residual does not vanish as $T \rightarrow \infty$, and asymptotically sits a factor of $N^{1/4}$ above the $\mathcal{O}(N^{-1/2})$ deployment floor of their own Theorem 1. Eliminating this residual is the direct motivation for our time-decaying schedule, which preserves the no-communication, no-oracle setting of Yardim et al. [2025].

Multi-player multi-armed bandits. Independent learning without communication has also been studied in MMAB with hard or soft collisions [Anandkumar et al., 2011, Avner and Mannor, 2014, Rosenski et al., 2016, Bubeck et al., 2021, Lugosi and Mehrabian, 2022]. While the no-communication aspect is shared, that line targets cooperative regret rather than equilibrium exploitability, and uses different algorithmic primitives such as musical-chairs schedules [Rosenkshi et al., 2016].

Mechanism design. Our principal-intervention interpretation connects to the mechanism-design literature on incentivizing decentralized populations [Roughgarden, 2016]; we provide a first zeroth-order bilevel formulation rather than a complete mechanism-design solution. The interaction between optimal mechanism design and decentralized population learning under sample-budget constraints remains an open problem.

Our position. In summary, existing approaches either rely on central coordination (classical VI, alternating MFG-RL) or accept a sub-optimal learning floor (Yardim’s TRPA with $N^{-1/4}$). Our work occupies the middle ground: achieving centralized-level convergence through purely decentralized, time-decaying intervention.

Paper organization. Section 2 introduces SMFG, the variational inequality formulation, and recalls TRPA-Full. Section 3 presents A-TRPA. Section 4 extends the framework to the bilevel Stackelberg-MFG setting and presents B-ATRPA. Section 5 contains the formal convergence analysis for A-TRPA. Appendix C gives a heuristic sufficient-condition sketch for B-ATRPA. Section 6 reports experimental results across three benchmarks. Section 7 discusses limitations and future work.

2 Preliminaries

2.1 Static Mean-Field Games and Nash Equilibrium

Consider a static mean-field game (SMFG) with N agents and a common finite action set $\mathcal{A}$ ($|\mathcal{A}| = K$). Let $\Delta_{\mathcal{A}}$ denote the probability simplex over $\mathcal{A}$, and let $e_a \in \mathbb{R}^K$ denote the canonical basis vector corresponding to action $a \in \mathcal{A}$. A *strategy profile* is a tuple $(\pi^1, \dots, \pi^N)$ with each $\pi^i \in \Delta_{\mathcal{A}}$. When the profile is played, agents independently sample $a^j \sim \pi^j$, yielding the realized *empirical action distribution*

$$\hat{\mu} = \frac{1}{N} \sum_{j=1}^N e_{a^j} \in \Delta_{\mathcal{A}}.$$

The payoff structure is given by an operator $\mathbf{F} : \Delta_{\mathcal{A}} \rightarrow [0, 1]^K$ that returns a payoff vector over actions; the scalar payoff associated with action $a \in \mathcal{A}$ under empirical distribution $\hat{\mu}$ is the entry $[\mathbf{F}(\hat{\mu})]_a$. Agent i ’s expected payoff

under the profile is therefore

$$V^i(\pi^1, \dots, \pi^N) := \mathbb{E}_{a^j \sim \pi^j, j=1, \dots, N} [\mathbf{F}(\hat{\mu})]_{a^i}, \quad \hat{\mu} = \frac{1}{N} \sum_{j=1}^N e_{a^j}. \quad (1)$$

The exploitability of agent i at the profile is

$$\mathcal{E}_{\text{exp}}^i(\pi^1, \dots, \pi^N) := \max_{\pi' \in \Delta_{\mathcal{A}}} V^i(\pi', \pi^{-i}) - V^i(\pi^1, \dots, \pi^N),$$

where π^{-i} denotes the profile of all agents other than i . A profile is a δ -Nash equilibrium (δ -NE) if $\max_{i \in [N]} \mathcal{E}_{\text{exp}}^i \leq \delta$.

Mean-Field Variational Inequality (MF-VI). In the infinite-agent limit, and under the payoff-maximization convention, the mean-field Nash equilibrium (MF-NE) π^* solves the variational inequality

$$\langle \mathbf{F}(\pi^*), \pi^* - \pi \rangle \geq 0, \quad \forall \pi \in \Delta_{\mathcal{A}}. \quad (2)$$

Yardim et al. [2025, Theorem 1] establish that homogeneous deployment of the exact MF-NE, $(\pi^*, \dots, \pi^*) \in \Delta_{\mathcal{A}}^N$, forms an $\mathcal{O}(1/\sqrt{N})$ -NE of the N -player game. This provides the finite-population homogeneous-deployment benchmark.

Assumptions. We adopt the standard SMFG regularity conditions from Yardim et al. [2025]:

Assumption 1 (Regularity and Noise). (1) **Lipschitz:** $\mathbf{F}$ is L -Lipschitz on $\Delta_{\mathcal{A}}$. (2) **Payoff monotonicity (Lasry–Lions, payoff-maximization convention):** $\langle \mathbf{F}(\mu_1) - \mathbf{F}(\mu_2), \mu_1 - \mu_2 \rangle \leq 0$ for all $\mu_1, \mu_2 \in \Delta_{\mathcal{A}}$ (equivalently, $-\mathbf{F}$ is monotone in the standard VI sense; this is the natural sign for crowd-aversion / congestion-type payoffs). (3) **Noise (full-feedback observation model):** In the full-feedback setting analyzed here, at round t agent i observes a noisy payoff vector $r_t^i = \mathbf{F}(\hat{\mu}_t) + n_t^i \in \mathbb{R}^K$, where $\hat{\mu}_t = \frac{1}{N} \sum_{j=1}^N e_{a_t^j}$ is the empirical distribution of the realized actions $\{a_t^j\}_{j=1}^N$ at round t . Letting $\mathcal{F}_t$ denote the natural filtration up to round t , the noise satisfies $\mathbb{E}[n_t^i | \mathcal{F}_t] = 0$, is independent across i given $\mathcal{F}_t$, and has entrywise conditional variance bounded by σ^2 .

TRPA-Full and the $N^{1/4}$ Gap. Yardim et al. [2025] propose *Tikhonov-Regularized Projected Ascent with full feedback* (TRPA-Full): each agent observes a noisy payoff vector r_t^i over all actions and updates its iterate $\pi_t^i \in \Delta_{\mathcal{A}}$ via a projected ascent step with a Tikhonov shrinkage term,

$$\pi_{t+1}^i = \Pi_{\Delta_{\mathcal{A}}}((1 - \eta_t \tau) \pi_t^i + \eta_t r_t^i),$$

where $\Pi_{\Delta_{\mathcal{A}}}$ is the Euclidean projection onto $\Delta_{\mathcal{A}}$ and $\tau > 0$ is the constant Tikhonov parameter. With this update, TRPA-Full achieves an exploitability bound of $\mathcal{O}(\tau^{-2}/\sqrt{T} + \tau^{-1}/\sqrt{N} + \tau)$. For fixed N , this residual does not vanish with additional learning time; optimizing the finite-population terms gives $\tau = N^{-1/4}$ and an $\mathcal{O}(N^{-1/4})$ asymptotic-in- T floor—a factor of $N^{1/4}$ above the $\mathcal{O}(N^{-1/2})$ deployment limit, and the gap we target with a time-decaying schedule.

3 The A-TRPA Algorithm

To close the aforementioned $N^{1/4}$ performance gap, we introduce Annealed-TRPA (A-TRPA). The single algorithmic modification is to replace the constant regularization with the time-decaying *annealed Tikhonov schedule* $\tau_t = \eta_t = t^{-1/2}$ ($\alpha = \beta = 1/2$, satisfying $\alpha + \beta = 1$). Each agent maintains the weighted averaged iterate $\bar{\pi}_T^i = S_T^{-1} \sum_t \eta_t \pi_t^i$; the theoretical guarantee (Theorem 4) is stated for the deterministic population-and-expectation average $\hat{\pi}_T = (1/N) \sum_i \mathbb{E}[\bar{\pi}_T^i]$, whose homogeneous deployment is an $\tilde{\mathcal{O}}(T^{-1/2} + N^{-1/2} + \sigma)$ -NE. Under noiseless feedback ($\sigma = 0$, full feedback) all agents are pathwise homogeneous so $\hat{\pi}_T = \mathbb{E}[\bar{\pi}_T^1]$, the rate becomes $\tilde{\mathcal{O}}(T^{-1/2} + N^{-1/2})$, matches the deployment-level floor of Theorem 1 of Yardim et al. [2025], and strictly improves the $N^{-1/4}$ rate of TRPA-Full. The realized last iterate empirically matches this rate (Section 6); a rigorous last-iterate analysis at the annealed Tikhonov schedule remains open.

Dynamic Principal Intervention. The A-TRPA update direction, $r_t^i - \tau_t \pi_t^i$, admits a mechanism-design reading. Each agent is implicitly maximizing an ℓ^2 -regularized objective:

$$\nabla_{\pi^i} \left(\langle r_t^i, \pi^i \rangle - \frac{\tau_t}{2} \|\pi^i\|_2^2 \right) = r_t^i - \tau_t \pi^i.$$

The ℓ^2 penalty $\frac{\tau_t}{2} \|\pi^i\|_2^2$ favors more mixed policies; in the absence of payoff signal, the regularized objective is maximized at the uniform policy. We interpret $-\tau_t \pi_t^i$ as a *dynamic intervention* imposed by an underlying principal: heavy regularization initially (high τ_t) prevents early divergence in the absence of communication, and the intervention gradually withdraws ($\tau_t \rightarrow 0$) as the population synchronizes, enabling unbiased convergence.

Algorithm 1 TRPA-Full [Yardim et al., 2025]

Require: Constant τ , stepsize $\{\eta_t\}$.

```

1: Init  $\pi_1^i \leftarrow \text{Unif}(\mathcal{A})$ .
2: for  $t = 1, \dots, T$  do
3:   Play  $a_t^i \sim \pi_t^i$ .
4:   Observe  $r_t^i = \mathbf{F}(\hat{\mu}_t) + n_t^i$ .
5:    $\pi_{t+1}^i \leftarrow \Pi_{\Delta_{\mathcal{A}}}((1 - \eta_t \tau) \pi_t^i + \eta_t r_t^i)$ .
6: end for
7: return  $\pi_T^i$ .
```

Algorithm 2 A-TRPA (ours)

Require: Annealed Tikhonov schedule

```

 $\tau_t = \eta_t = t^{-1/2}$ .
1: Init  $\pi_1^i \leftarrow \text{Unif}(\mathcal{A})$ .
2: for  $t = 1, \dots, T$  do
3:   Play  $a_t^i \sim \pi_t^i$ .
4:   Observe  $r_t^i = \mathbf{F}(\hat{\mu}_t) + n_t^i$ .
5:    $\pi_{t+1}^i \leftarrow \Pi_{\Delta_{\mathcal{A}}}((1 - \eta_t \tau_t) \pi_t^i + \eta_t r_t^i)$ .
6:   Update average  $\bar{\pi}_t^i \leftarrow S_t^{-1} \sum_{s \leq t} \eta_s \pi_s^i$ .
7: end for
8: return  $\bar{\pi}_T^i$   $\triangleright$  averaged iterate;  $\pi_T^i$  tracked as empirical diagnostic
```

Algorithm 3 B-ATRP (ours, bilevel)

Require: $H, T, \delta, \{\eta_h^{\text{out}}\}$.

```

1: Init  $w_0 \in W$ .
2: for  $h = 0, \dots, H-1$  do
3:    $u_h \sim \text{Unif}(\mathbb{S}^{d-1})$ .
4:    $w_h^\pm \leftarrow w_h \pm \delta u_h$ .
5:    $\bar{\pi}^\pm \leftarrow \text{A-TRPA}(\mathbf{F}(\cdot; w_h^\pm), T)$   $\triangleright$  inner returns averaged iterate  $\bar{\pi}_T^\pm$ 
6:    $\hat{U}^\pm \leftarrow \mathcal{U}(\bar{\pi}^\pm, w_h^\pm)$ .
7:    $\hat{g}_h \leftarrow \frac{d}{2\delta} (\hat{U}^+ - \hat{U}^-) u_h$ .
8:    $w_{h+1} \leftarrow \Pi_W(w_h + \eta_h^{\text{out}} \hat{g}_h)$ .
9: end for
10: return  $w_H$ .
```

Figure 1: TRPA-Full and A-TRPA are population-level learning algorithms for a fixed SMFG. B-ATRP is a bilevel extension that uses A-TRPA as an inner learner for a principal-level zeroth-order update. **TRPA-Full** (left) uses constant τ and returns the last iterate, attaining the $\mathcal{O}(N^{-1/4})$ floor. **A-TRPA** (middle, ours) uses the annealed Tikhonov schedule $\tau_t = \eta_t = t^{-1/2}$ and returns the weighted averaged iterate $\bar{\pi}_T^i$ as its primary output. Theorem 4 bounds the homogeneous-deployment exploitability of the deterministic average $\bar{\pi}_T = (1/N) \sum_i \mathbb{E}[\bar{\pi}_T^i]$ at $\tilde{\mathcal{O}}(T^{-1/2} + N^{-1/2} + \sigma)$; we additionally track the last iterate π_T^i as an empirical diagnostic, which tracks the same rate (the missing last-iterate guarantee is discussed in Appendix B). **B-ATRP** (right, ours) embeds A-TRPA inside a zeroth-order projected ascent at the principal level, computing $\bar{\pi}_T^\pm$ for two perturbed payoffs $\mathbf{F}(\cdot; w_h^\pm)$ to estimate the welfare gradient.

4 From Principal Intervention to Bilevel Stackelberg-MFG

The principal interpretation in Section 3 views the Tikhonov term as a *passive* centralized intervention—following a prescribed mathematical schedule to stabilize learning. We now upgrade the framework to an *active* principal that strategically chooses a policy to maximize a system-level welfare objective, while the population still learns independently via A-TRPA. We propose a bilevel extension tailored to decentralized SMFG learning: the principal optimizes via zeroth-order welfare evaluations while agents continue to run A-TRPA without communication.

4.1 Stackelberg-MFG Setup

Let $W \subset \mathbb{R}^d$ be a compact convex set of principal actions (e.g., tolls, subsidies, capacity allocations). Let $\mathbf{F} : \Delta_{\mathcal{A}} \times W \rightarrow [0, 1]^K$ be a parametric family of payoff operators, with $\mathbf{F}(\cdot; w)$ playing the role of the SMFG payoff when the principal commits to $w \in W$. We make the following structural assumptions:

Assumption 2 (Uniform regularity). *For every $w \in W$, $\mathbf{F}(\cdot; w)$ is L -Lipschitz, and $-\mathbf{F}(\cdot; w)$ is λ -strongly monotone in its first argument uniformly in w ; equivalently, in the payoff-maximization convention,*

$$\langle \mathbf{F}(\mu; w) - \mathbf{F}(\nu; w), \mu - \nu \rangle \leq -\lambda \|\mu - \nu\|_2^2 \quad \forall \mu, \nu \in \Delta_{\mathcal{A}}.$$

Moreover, $\mathbf{F}$ is L_w -Lipschitz in w uniformly in μ , that is, $\|\mathbf{F}(\mu; w) - \mathbf{F}(\mu; w')\|_2 \leq L_w \|w - w'\|_2$.

Under Assumption 2, the MF-NE map $w \mapsto \pi^*(w)$ (the unique solution of the parametric VI $\langle \mathbf{F}(\pi^*(w); w), \pi^*(w) - \pi \rangle \geq 0$ for all $\pi \in \Delta_{\mathcal{A}}$) is well-defined, and standard VI sensitivity bounds give $\|\pi^*(w) - \pi^*(w')\|_2 \leq (L_w/\lambda) \|w - w'\|_2$.

The principal seeks to maximize a welfare functional $U : W \rightarrow \mathbb{R}$ that depends on both the intervention w and the equilibrium response $\pi^*(w)$ it induces:

$$U(w) := \mathcal{U}(\pi^*(w), w), \tag{3}$$

where $\mathcal{U} : \Delta_{\mathcal{A}} \times W \rightarrow \mathbb{R}$ is a L_U -Lipschitz, β -smooth functional. Concrete examples include:

- **Social welfare:** $\mathcal{U}(\pi, w) = \pi^\top \mathbf{F}(\pi; w)$ (population’s total expected payoff at equilibrium).
- **Negative congestion:** $\mathcal{U}(\pi, w) = -\pi^\top D(\pi)$ for delay function D (e.g. traffic tolling).
- **Revenue:** $\mathcal{U}(\pi, w) = w^\top \pi$ minus an equilibrium-dependent cost (e.g. pricing under demand response).

Information structure. The principal observes only noisy welfare evaluations $\hat{U}(w) = U(w) + \xi$ with $\mathbb{E}[\xi] = 0$ and $|\xi| \leq \sigma_U$. In particular, the principal does not have access to $\mathbf{F}$, π^* , or individual agent payoffs. Agents remain decentralized: each observes only its local noisy full-feedback payoff vector as in Section 3 and does not communicate with other agents.

4.2 Bilevel A-TRPA Algorithm

We propose a two-loop scheme that nests A-TRPA inside a zeroth-order projected-ascent update on w .

The estimator $\hat{g}_h = \frac{d}{2\delta}(\hat{U}^+ - \hat{U}^-)u_h$ is the standard two-point symmetric Flaxman–Kalai–McMahan zeroth-order estimator [Flaxman et al., 2005] for the sphere-smoothed welfare $U_\delta(w) := \mathbb{E}_{u \sim \mathbb{S}}[U(w + \delta u)]$. Two features distinguish our setting from the classical zeroth-order literature: (i) each evaluation $\hat{U}^\pm$ is itself the output of an inner-level learning algorithm (A-TRPA), introducing systematic inner-loop bias on top of the smoothing bias; (ii) agents are finite in number and decentralized, contributing finite-population exploitability error.

4.3 Principal Intervention as a Unifying Framework

The two contributions of this paper – A-TRPA (Section 3) and B-ATRPA (this section) – are not two separate ideas but two operating points on a single design axis: *the strength of principal intervention*. We make this explicit through three regimes of increasing intervention.

Regime 0 (no principal): TRPA-Full at fixed τ . Yardim et al. [2025] analyze independent learning with constant Tikhonov regularization. The schedule is hand-tuned by the algorithm designer to minimize the asymptotic exploitability bound, but no agency is exercised over the dynamics. This achieves the $\mathcal{O}(N^{-1/4})$ floor.

Regime 1 (passive principal): A-TRPA. We reinterpret the regularizer $-\tau_t \pi_t^i$ as a centralized intervention applied uniformly to all agents. The principal here is *passive*: it commits ex ante to a schedule $\tau_t = t^{-1/2}$ and exerts no strategic choice during learning, only modulating the implicit exploration–exploitation tradeoff over time. This single change already closes the $N^{1/4}$ gap, recovering the deployment-level $N^{-1/2}$ floor (Theorem 4).

Regime 2 (active principal): B-ATRPA. We let the principal optimize a separate strategic variable $w \in W$ to maximize a system-level welfare functional $U(w)$, while the population continues to learn via A-TRPA at the previously-fixed τ_t schedule. The principal observes only noisy welfare evaluations $\hat{U}(w)$ and runs zeroth-order Flaxman–Kalai–McMahan ascent on the smoothed objective U_δ . The two timescales – inner population learning ($t = 1, \dots, T$) and outer principal updates ($h = 0, \dots, H-1$) – are orthogonal: τ_t governs the inner regularizer, w_h governs the payoff itself. An ℓ^2 -regularized welfare $\mathcal{U}(\pi, w) - \tau \|w\|^2$ makes the inner/outer analogy literal: both regimes are Tikhonov regularization, applied at different layers.

This three-regime view positions our contribution sharply: Yardim et al. [2025] solved the no-principal case; A-TRPA introduces the passive principal and closes the $N^{1/4}$ gap; B-ATRPA extends this viewpoint to an active-principal setting, which we study algorithmically and empirically. A complete outer-loop convergence theory is left for future work. The unifying conceptual move is to recognize that the regularizer in TRPA was already an implicit principal – and that making this principal explicit, then progressively granting it agency, yields a coherent two-step research program.

5 Convergence Analysis

This section establishes the main formal guarantee for A-TRPA: a homogeneous-deployment exploitability bound for the deterministic averaged iterate $\hat{\pi}_T$ at the annealed Tikhonov schedule $\tau_t = \eta_t = t^{-1/2}$. The last-iterate behaviour of A-TRPA is treated empirically in Section 6; a heuristic sufficient-condition discussion is deferred to Appendix B, and a sufficient-condition sketch toward a B-ATRPA non-asymptotic outer-loop guarantee is deferred to Appendix C.

5.1 Tools

We isolate three ingredients used in the main proof: a finite-population bias bound for the mean-field operator (Lemma 1), a consensus bound across agents (Lemma 2), and a Minty-to-exploitability conversion (Lemma 3). Lemma 1 requires the Jacobian of $\mathbf{F}$ to be Lipschitz; this strengthens the operator regularity beyond Assumption 1. This condition holds for the smooth Linear and KL benchmarks, but not for the nonsmooth Beach-Bar benchmark; BB is therefore used as an out-of-scope robustness test.

Assumption 3 (Jacobian-Lipschitz payoff operator). $\mathbf{F} : \Delta_{\mathcal{A}} \rightarrow \mathbb{R}^K$ is continuously differentiable, and its Jacobian is L_H -Lipschitz: for all $x, y \in \Delta_{\mathcal{A}}$,

$$\|\nabla \mathbf{F}(x) - \nabla \mathbf{F}(y)\|_2 \leq L_H \|x - y\|_2.$$

Equivalently, $\|\mathbf{F}(x) - \mathbf{F}(y) - \nabla \mathbf{F}(y)(x - y)\|_2 \leq \frac{L_H}{2} \|x - y\|_2^2$.

Lemma 1. Let $\xi_t := F(\hat{\mu}_t) - F(\bar{\mu}_t)$, $\bar{\mu}_t := \frac{1}{N} \sum_{j=1}^N \pi_t^j$. Under the previous assumption, we have :

$$\|\mathbb{E}[\xi_t \mid \mathcal{F}_t]\|_2 \leq \frac{L_H}{2N}.$$

Moreover,

$$\mathbb{E}[\|\xi_t - \mathbb{E}[\xi_t \mid \mathcal{F}_t]\|_2^2 \mid \mathcal{F}_t] \leq \frac{L^2}{N}.$$

So, for any $\mathcal{F}_t$ -measurable vector $v_t \in \mathbb{R}^K$ and any $\delta \in (0, 1)$,

$$2\eta_t \mathbb{E}[\langle v_t, \xi_t \rangle] \leq \eta_t \tau_t \delta \mathbb{E}[\|v_t\|_2^2] + \frac{\eta_t L_H^2}{4\tau_t \delta N^2}.$$

Proof. Conditionally on $\mathcal{F}_t$, the random variables $a_t^1, \dots, a_t^N$ are independent with $a_t^j \sim \pi_t^j$. Having $\hat{\mu}_t = \frac{1}{N} \sum_{j=1}^N e_{a_t^j}$, we can deduce that :

$$\mathbb{E}[\hat{\mu}_t \mid \mathcal{F}_t] = \mathbb{E}\left[\frac{1}{N} \sum_{i=1}^N e_{a_t^i} \mid \mathcal{F}_t\right] = \frac{1}{N} \sum_{i=1}^N \mathbb{E}[e_{a_t^i} \mid \mathcal{F}_t] = \frac{1}{N} \sum_{i=1}^N \sum_{a \in \mathcal{A}} \pi_t^i(a) e_a = \frac{1}{N} \sum_{j=1}^N \pi_t^j = \bar{\mu}_t$$

So (Eq.1) : $\mathbb{E}[\hat{\mu}_t - \bar{\mu}_t \mid \mathcal{F}_t] = \mathbb{E}[\hat{\mu}_t \mid \mathcal{F}_t] - \mathbb{E}[\bar{\mu}_t \mid \mathcal{F}_t] = 0$

By the assumption, we can do the Taylor expansion for $F(\hat{\mu}_t)$ on $\bar{\mu}_t$:

$$F(\hat{\mu}_t) = F(\bar{\mu}_t) + \nabla F(\bar{\mu}_t)(\hat{\mu}_t - \bar{\mu}_t) + R_t$$

where $R_t = F(\hat{\mu}_t) - F(\bar{\mu}_t) - \nabla F(\bar{\mu}_t)(\hat{\mu}_t - \bar{\mu}_t)$ is by assumption $\|R_t\|_2 \leq \frac{L_H}{2} \|\hat{\mu}_t - \bar{\mu}_t\|_2^2$. This equation gives us :

$$F(\hat{\mu}_t) - F(\bar{\mu}_t) = \nabla F(\bar{\mu}_t)(\hat{\mu}_t - \bar{\mu}_t) + R_t$$

Taking conditional expectation, the first-order term vanishes by (Eq.1):

$$\mathbb{E}[F(\hat{\mu}_t) - F(\bar{\mu}_t) \mid \mathcal{F}_t] = \nabla F(\bar{\mu}_t) \mathbb{E}[\hat{\mu}_t - \bar{\mu}_t \mid \mathcal{F}_t] + \mathbb{E}[R_t \mid \mathcal{F}_t] = \mathbb{E}[R_t \mid \mathcal{F}_t].$$

Thus

$$\|\mathbb{E}[F(\hat{\mu}_t) - F(\bar{\mu}_t) \mid \mathcal{F}_t]\|_2 = \|\mathbb{E}[R_t \mid \mathcal{F}_t]\|_2 \leq \frac{L_H}{2} \mathbb{E}[\|\hat{\mu}_t - \bar{\mu}_t\|_2^2 \mid \mathcal{F}_t].$$

Recall that $\bar{\mu}_t := \frac{1}{N} \sum_{i=1}^N \pi_t^i$. So $\hat{\mu}_t - \bar{\mu}_t = \frac{1}{N} \sum_{j=1}^N (e_{a_t^j} - \pi_t^j)$. Then one can develop the norm, knowing $\|e_a\|_2 = 1$ to finally deduce that :

$$\mathbb{E}[\|\hat{\mu}_t - \bar{\mu}_t\|_2^2 \mid \mathcal{F}_t] = \frac{1}{N^2} \sum_{j=1}^N \mathbb{E}[\|e_{a_t^j} - \pi_t^j\|_2^2 \mid \mathcal{F}_t] = \frac{1}{N^2} \sum_{j=1}^N \underbrace{1 - \|\pi_t^j\|_2^2}_{\leq 1} \leq \frac{1}{N}$$

This proves our first inequality :

$$\|\mathbb{E}[F(\hat{\mu}_t) - F(\bar{\mu}_t) \mid \mathcal{F}_t]\|_2 \leq \frac{L_H}{2N}.$$

The second inequality is easy to prove, we know that for any random variable Z we have $\mathbb{E}[\|Z - \mathbb{E}[Z]\|^2] = \mathbb{E}[\|Z\|^2] - \|\mathbb{E}[Z]\|^2 \leq \mathbb{E}[\|Z\|^2]$. So we let $Z = \xi_t = F(\hat{\mu}_t) - F(\bar{\mu}_t)$ and conditioning by $\mathcal{F}_t$, we have :

$$\mathbb{E}[\|\xi_t - \mathbb{E}[\xi_t \mid \mathcal{F}_t]\|_2^2 \mid \mathcal{F}_t] \leq \mathbb{E}[\|\xi_t\|_2^2 \mid \mathcal{F}_t] \leq \frac{L^2}{N}$$

Because F is Lipschitz (i.e $\|F(\hat{\mu}_t) - F(\bar{\mu}_t)\|_2 \leq L \|\hat{\mu}_t - \bar{\mu}_t\|_2$), and the inequality remains with expectation :

$$\mathbb{E}[\|F(\hat{\mu}_t) - F(\bar{\mu}_t)\|_2^2 \mid \mathcal{F}_t] \leq L^2 \underbrace{\mathbb{E}[\|\hat{\mu}_t - \bar{\mu}_t\|_2^2 \mid \mathcal{F}_t]}_{\leq 1/N, \text{ as proved}} \leq \frac{L^2}{N}.$$

The third inequality can be start with :

$$2\eta_t \mathbb{E}[\langle v_t, \xi_t \rangle \mid \mathcal{F}_t] = 2\eta_t \langle v_t, \mathbb{E}[\xi_t \mid \mathcal{F}_t] \rangle$$

Which by Cauchy, then proved $\|\mathbb{E}[\xi_t \mid \mathcal{F}_t]\|_2 \leq \frac{L_H}{2N}$:

$$2\eta_t \langle v_t, \mathbb{E}[\xi_t \mid \mathcal{F}_t] \rangle \leq 2\eta_t \|v_t\|_2 \|\mathbb{E}[\xi_t \mid \mathcal{F}_t]\|_2 \leq 2\eta_t \|v_t\|_2 \frac{L_H}{2N}$$

Finally we can use Young inequality as done in the paper to conclude. $\square$

Lemma 2 (Consensus across agents under noisy feedback). *Under Assumption 1, for any $\eta_t \tau_t \in [0, 1]$ and any $i \neq j \in [N]$,*

$$\mathbb{E}[\|\pi_{t+1}^i - \pi_{t+1}^j\|_2^2] \leq (1 - \eta_t \tau_t) \mathbb{E}[\|\pi_t^i - \pi_t^j\|_2^2] + 2K\sigma^2 \eta_t^2.$$

At the annealed Tikhonov schedule $\tau_t = \eta_t = t^{-1/2}$ with the common uniform initialization $\pi_1^i = \text{Unif}(\mathcal{A})$, unrolling the recursion gives, for every agent $i \in [N]$ and every $t \geq 1$,

$$\mathbb{E}[e_t^i] \leq 2K\sigma^2, \quad e_t^i := \|\pi_t^i - \bar{\mu}_t\|_2^2.$$

Proof. One-step recursion. For agents $i \neq j$, subtracting the A-TRPA updates and using non-expansiveness of $\Pi_{\Delta_{\mathcal{A}}}$,

$$\|\pi_{t+1}^i - \pi_{t+1}^j\|_2^2 \leq \|(1 - \eta_t \tau_t)(\pi_t^i - \pi_t^j) + \eta_t(r_t^i - r_t^j)\|_2^2.$$

Both agents observe the same empirical distribution $\hat{\mu}_t$, so the deterministic part of r_t^i cancels and $r_t^i - r_t^j = n_t^i - n_t^j$. Squaring and taking conditional expectation given $\mathcal{F}_t$, the cross term $2\eta_t(1 - \eta_t \tau_t)\langle \pi_t^i - \pi_t^j, n_t^i - n_t^j \rangle$ has zero conditional mean by Assumption 1, item (3), and $\mathbb{E}[\|n_t^i - n_t^j\|_2^2 \mid \mathcal{F}_t] \leq 2K\sigma^2$. Hence

$$\mathbb{E}[\|\pi_{t+1}^i - \pi_{t+1}^j\|_2^2 \mid \mathcal{F}_t] \leq (1 - \eta_t \tau_t)^2 \|\pi_t^i - \pi_t^j\|_2^2 + 2K\sigma^2 \eta_t^2.$$

Using $(1 - \eta_t \tau_t)^2 \leq 1 - \eta_t \tau_t$ for $\eta_t \tau_t \in [0, 1]$ and taking total expectation gives the stated recursion.

Unrolling at the annealed Tikhonov schedule. Set $D_t := \mathbb{E}[\|\pi_t^i - \pi_t^j\|_2^2]$. For $\tau_t = \eta_t = t^{-1/2}$, $\eta_t \tau_t = 1/t$ and $\eta_t^2 = 1/t$, so

$$D_{t+1} \leq \left(1 - \frac{1}{t}\right) D_t + \frac{2K\sigma^2}{t}, \quad \text{equivalently} \quad t D_{t+1} \leq (t-1) D_t + 2K\sigma^2.$$

Let $\phi_t := (t-1) D_t$, so $\phi_{t+1} = t D_{t+1}$ and the recursion reads $\phi_{t+1} \leq \phi_t + 2K\sigma^2$. The common uniform initialization gives $D_1 = 0$, hence $\phi_1 = 0$. Telescoping from $s = 1$ to t yields $\phi_{t+1} \leq 2K\sigma^2 \cdot t$, hence $D_{t+1} \leq 2K\sigma^2$ for every $t \geq 1$.

Bound on e_t^i . By Jensen's inequality applied to $\|\cdot\|_2^2$,

$$e_t^i = \left\| \frac{1}{N} \sum_{j=1}^N (\pi_t^i - \pi_t^j) \right\|_2^2 \leq \frac{1}{N} \sum_{j=1}^N \|\pi_t^i - \pi_t^j\|_2^2,$$

so $\mathbb{E}[e_t^i] \leq (1/N) \sum_j D_t \leq 2K\sigma^2$. $\square$

Lemma 3 (Minty gap to homogeneous-deployment exploitability). *We use exactly the Minty-gap convention of Yardim et al. [2025], rewritten under our payoff-maximization sign convention. Let Assumption 1 hold and let $\bar{\pi} \in \Delta_{\mathcal{A}}$ be a deterministic policy with*

$$\sup_{\pi \in \Delta_{\mathcal{A}}} \langle \mathbf{F}(\pi), \pi - \bar{\pi} \rangle \leq \varepsilon.$$

Then the homogeneous N -player profile $(\bar{\pi}, \dots, \bar{\pi})$ satisfies

$$\max_{i \in [N]} \mathcal{E}_{\text{exp}}^i(\bar{\pi}, \dots, \bar{\pi}) \leq M(K, L) \varepsilon + \mathcal{O}(N^{-1/2}),$$

where $M(K, L)$ depends only on $K = |\mathcal{A}|$ and the Lipschitz modulus L of $\mathbf{F}$.

Source. The bound is a quoted composition of two results in Yardim et al. [2025], restated under our payoff-maximization sign convention. Their Lemma 2 shows that under Assumption 1 a Minty gap of size ε implies that $\bar{\pi}$ is an $M(K, L)\varepsilon$ -MF-NE, with constant $M(K, L)$ depending only on K and L . Their Theorem 1 then provides the additive $\mathcal{O}(N^{-1/2})$ finite-population sampling penalty when an MF-NE is deployed homogeneously across N agents. Composition gives the stated bound. We use these as black-box conversions and do not re-derive them here. $\square$

5.2 Main Convergence Theorem

We analyze the **weighted averaged iterate**

$$\bar{\pi}_T^i := \frac{1}{S_T} \sum_{t=1}^T \eta_t \pi_t^i, \quad S_T := \sum_{t=1}^T \eta_t,$$

and its deterministic population-and-expectation average $\hat{\pi}_T := (1/N) \sum_i \mathbb{E}[\bar{\pi}_T^i] \in \Delta_{\mathcal{A}}$. Averaged-iterate VI-gap analysis is the standard route to closed rates for monotone variational inequalities [Nemirovski, 2004, Kotsalis et al., 2022]: telescoping eliminates the recursive coupling, and Jensen's inequality with monotonicity of $\mathbf{F}$ moves the operator outside the time sum, turning the per-step $\mathcal{O}(1/t)$ variance source into a $\tilde{\mathcal{O}}(T^{-1/2})$ contribution that vanishes in T . The output of this argument is a bound on the Minty (dual) gap of $\hat{\pi}_T$, which Lemma 3 (proved in Section 5.1) converts into homogeneous-deployment exploitability.

Theorem 4 (A-TRPA averaged-iterate convergence). *Let Assumption 1 (regularity and noise) and Assumption 3 (Jacobian-Lipschitz $\mathbf{F}$) hold. Define the deterministic population-and-expectation averaged policy*

$$\hat{\pi}_T := \frac{1}{N} \sum_{i=1}^N \mathbb{E}[\bar{\pi}_T^i] \in \Delta_{\mathcal{A}}.$$

Under the annealed Tikhonov schedule $\tau_t = \eta_t = t^{-1/2}$, $\hat{\pi}_T$ satisfies the deterministic Minty gap bound

$$\sup_{\pi \in \Delta_{\mathcal{A}}} \langle \mathbf{F}(\pi), \pi - \hat{\pi}_T \rangle \leq \tilde{\mathcal{O}}\left(T^{-1/2} + \frac{1}{N} + \sigma\right), \quad (4)$$

and consequently the homogeneous deployment $(\hat{\pi}_T, \dots, \hat{\pi}_T)$ is an

$$\tilde{\mathcal{O}}\left(T^{-1/2} + \frac{1}{\sqrt{N}} + \sigma\right)$$

-Nash equilibrium of the N -player game, where $\tilde{\mathcal{O}}$ absorbs $\log T$ factors. Under noiseless feedback ($\sigma = 0$), the bound becomes $\tilde{\mathcal{O}}(T^{-1/2} + N^{-1/2})$, matching the deployment-level floor of Theorem 1 of Yardim et al. [2025] and closing the $N^{1/4}$ gap of the constant- τ bound (Corollary 1 of Yardim et al. 2025).

Remark 1 (On the deterministic policy $\hat{\pi}_T$). *Three observations. (i) Under noiseless feedback ($\sigma = 0$, full feedback), all agents observe the same payoff vector $r_t = \mathbf{F}(\hat{\mu}_t)$ at every round. Starting from the common uniform initialization, the iterates remain pathwise homogeneous: $\bar{\pi}_T^i = \bar{\pi}_T^1$ for all i , so $\hat{\pi}_T = \mathbb{E}[\bar{\pi}_T^1]$ collapses to the (single-agent, single-policy) seed-average. (ii) Under noisy feedback ($\sigma > 0$), the agents' iterates are heterogeneous, and the N -player exploitability of the realized profile $\{\bar{\pi}_T^j\}_j$ requires a separate heterogeneous-to-homogeneous conversion that we do not perform here; Theorem 4 controls the homogeneous deployment of the deterministic average $\hat{\pi}_T$, not of the realized heterogeneous profile. (iii) The bound (4) is on a deterministic policy and a deterministic gap, sidestepping the standard order-of-sup issue $\mathbb{E}[\sup_{\pi} X(\pi)] \geq \sup_{\pi} \mathbb{E}[X(\pi)]$; relating the realized exploitability $\mathcal{E}_{\text{exp}}^i(\bar{\pi}_T^i)$ (reported in experiments) to $\sup_{\pi} \langle \mathbf{F}(\pi), \pi - \hat{\pi}_T \rangle$ requires concentration of $\bar{\pi}_T^i$ around $\hat{\pi}_T$, which we do not formalize and which is consistent with the empirical seed variability reported in Section 6.*

Proof. Fix any reference policy $\pi \in \Delta_{\mathcal{A}}$. By non-expansiveness of $\Pi_{\Delta_{\mathcal{A}}}$ applied to the A-TRPA update,

$$\|\pi_{t+1}^i - \pi\|_2^2 \leq \|\pi_t^i - \pi\|_2^2 + 2\eta_t \langle r_t^i - \tau_t \pi_t^i, \pi_t^i - \pi \rangle + \eta_t^2 \|r_t^i - \tau_t \pi_t^i\|_2^2.$$

Substituting $r_t^i = \mathbf{F}(\hat{\mu}_t) + n_t^i$ and rearranging, the noise inner-product $\mathbb{E}[\langle n_t^i, \pi_t^i - \pi \rangle] = 0$ vanishes since π_t^i is $\mathcal{F}_t$ -measurable and n_t^i has mean zero (Assumption 1, item (3)). Taking total expectation,

$$2\eta_t \mathbb{E} \langle \mathbf{F}(\hat{\mu}_t), \pi - \pi_t^i \rangle \leq \mathbb{E} \|\pi_t^i - \pi\|_2^2 - \mathbb{E} \|\pi_{t+1}^i - \pi\|_2^2 + C \eta_t^2 + 2\eta_t \tau_t, \quad (5)$$

where $C = \mathcal{O}(K(1 + \sigma^2) + (L + \tau_1)^2)$ uniformly bounds $\mathbb{E} \|r_t^i - \tau_t \pi_t^i\|_2^2$ on the simplex.

Step 1 (Operator decomposition). Insert and subtract $\mathbf{F}(\bar{\mu}_t)$ and $\mathbf{F}(\pi_t^i)$: $\langle \mathbf{F}(\hat{\mu}_t), \pi - \pi_t^i \rangle = \langle \mathbf{F}(\pi_t^i), \pi - \pi_t^i \rangle + A_t + C_t$, where $A_t := \langle \mathbf{F}(\hat{\mu}_t) - \mathbf{F}(\bar{\mu}_t), \pi - \pi_t^i \rangle$ and $C_t := \langle \mathbf{F}(\bar{\mu}_t) - \mathbf{F}(\pi_t^i), \pi - \pi_t^i \rangle$. By Lemma 1, $|\mathbb{E} A_t| \leq \mathcal{O}(L_H/N)$. By Lemma 2, at the annealed schedule $\mathbb{E}[e_t^i] = \mathcal{O}(\sigma^2)$, hence by Lipschitzness of $\mathbf{F}$, $|\mathbb{E} C_t| = \mathcal{O}(\sigma)$.

Step 2 (Telescoping). Summing (5) from $t = 1$ to T and using $\mathbb{E} \|\pi_{T+1}^i - \pi\|_2^2 \geq 0$,

$$\sum_{t=1}^T \eta_t \mathbb{E} \langle \mathbf{F}(\pi_t^i), \pi - \pi_t^i \rangle \leq \mathcal{O}(1) + C \sum_t \eta_t^2 + \sum_t \eta_t \tau_t + \mathcal{O}\left(\frac{1}{N}\right) \sum_t \eta_t + \mathcal{O}(\sigma) \sum_t \eta_t \sqrt{\eta_t / \tau_t}. \quad (6)$$

Step 3 (Jensen + monotonicity). By Assumption 1 item (2) (monotonicity of $\mathbf{F}$), $\langle \mathbf{F}(\pi_t^i), \pi - \pi_t^i \rangle \geq \langle \mathbf{F}(\pi), \pi - \pi_t^i \rangle$. Multiplying by $\eta_t \geq 0$, summing, and dividing by S_T , the right-hand side becomes $\langle \mathbf{F}(\pi), \pi - \bar{\pi}_T^i \rangle$ by linearity. Thus

$$\mathbb{E} \langle \mathbf{F}(\pi), \pi - \bar{\pi}_T^i \rangle \leq \frac{1}{S_T} \left[\mathcal{O}(1) + C \sum_t \eta_t^2 + \sum_t \eta_t \tau_t + \mathcal{O}(1/N) S_T + \mathcal{O}(\sigma) \sum_t \eta_t \sqrt{\eta_t / \tau_t} \right]. \quad (7)$$

Step 4 (Sum estimates at $\tau_t = \eta_t = t^{-1/2}$). $S_T = \sum_t t^{-1/2} = \Theta(\sqrt{T})$. $\sum_t \eta_t^2 = \sum_t 1/t = \Theta(\log T)$, so $\sum_t \eta_t^2 / S_T = \tilde{\mathcal{O}}(T^{-1/2})$. $\sum_t \eta_t \tau_t = \sum_t 1/t = \Theta(\log T)$, so $\sum_t \eta_t \tau_t / S_T = \tilde{\mathcal{O}}(T^{-1/2})$. At the boundary $\eta_t / \tau_t = 1$, so $\sum_t \eta_t \sqrt{\eta_t / \tau_t} = \sum_t \eta_t = S_T$, giving $\sum_t \eta_t \sqrt{\eta_t / \tau_t} / S_T = 1$ (irreducible σ -floor at the boundary).

Substituting into (7):

$$\mathbb{E} \langle \mathbf{F}(\pi), \pi - \bar{\pi}_T^i \rangle \leq \tilde{\mathcal{O}} \left(T^{-1/2} + \frac{1}{N} + \sigma \right). \quad (8)$$

Step 5 (From Minty gap to exploitability). By linearity of expectation, $\mathbb{E} \langle \mathbf{F}(\pi), \pi - \bar{\pi}_T^i \rangle = \langle \mathbf{F}(\pi), \pi - \mathbb{E}[\bar{\pi}_T^i] \rangle$ since $\mathbf{F}(\pi)$ is deterministic. Hence (8) becomes the *deterministic* pointwise bound

$$\langle \mathbf{F}(\pi), \pi - \mathbb{E}[\bar{\pi}_T^i] \rangle \leq \tilde{\mathcal{O}} \left(T^{-1/2} + \frac{1}{N} + \sigma \right), \quad \text{for every } \pi \in \Delta_{\mathcal{A}}, i \in [N].$$

Averaging over $i = 1, \dots, N$ and using $\hat{\pi}_T = (1/N) \sum_i \mathbb{E}[\bar{\pi}_T^i]$:

$$\langle \mathbf{F}(\pi), \pi - \hat{\pi}_T \rangle \leq \tilde{\mathcal{O}} \left(T^{-1/2} + \frac{1}{N} + \sigma \right), \quad \forall \pi \in \Delta_{\mathcal{A}}.$$

Both sides are now deterministic and the right-hand side is uniform in π , so we can take $\sup_{\pi}$ on the left without an order-of-sup issue, yielding the bound (4) of the theorem statement. Applying Lemma 3 to the deterministic policy $\hat{\pi}_T$ converts the Minty gap into homogeneous-deployment exploitability with an additive $\mathcal{O}(N^{-1/2})$ sampling penalty that dominates the $\mathcal{O}(1/N)$ residual. $\square$

Remark 2 (The σ -floor at the boundary). *The σ term in Theorem 4 is irreducible at the annealed Tikhonov schedule $\eta_t / \tau_t = 1$: the consensus contribution $\sum_t \eta_t \sqrt{\eta_t / \tau_t} / S_T$ equals 1 when $\eta_t = \tau_t$. At noiseless feedback ($\sigma = 0$) this floor vanishes and the bound becomes $\tilde{\mathcal{O}}(T^{-1/2} + N^{-1/2})$. Under the strict adaptive schedule $\beta > \alpha$ of Yardim et al. [2025] (e.g., $\tau_t = t^{-1/3}$, $\eta_t = t^{-2/3}$), the σ -floor is replaced by the vanishing $\sigma T^{-(\beta-\alpha)/2} = \sigma T^{-1/6}$ at the cost of a slower $T^{-1/3}$ overall rate.*

Remark 3 (Closing the $N^{1/4}$ gap). *The $N^{-1/2}$ term in Theorem 4 matches the deployment-level $N^{-1/2}$ floor of Theorem 1 of Yardim et al. [2025], strictly improving the $N^{-1/4}$ rate of TRPA-Full at constant τ (Corollary 1 of Yardim et al. 2025).*

Remark 4 (Last iterate: empirical only). *Theorem 4 concerns the deterministic averaged policy $\hat{\pi}_T$. The realized last iterate π_T^i empirically tracks the same $\tilde{\mathcal{O}}(T^{-1/2} + N^{-1/2})$ rate (Section 6). We do not prove a last-iterate bound. A heuristic sufficient condition (co-coercivity), an explicit identification of the missing residual-to-distance step under the simplex projection, and the moving-target drift recursion are deferred to Appendix B.*

Remark 5 (B-ATRP: an empirical extension). *B-ATRP (Section 4) is presented as an algorithmic extension and is evaluated empirically on Stackelberg-Linear in Section 6.8. A complete non-asymptotic outer-loop convergence proof is left for future work; a sufficient-condition sketch identifying when standard non-convex zeroth-order analysis yields a $\tilde{\mathcal{O}}(\epsilon^{-12}N)$ budget toward ϵ -stationarity of the smoothed welfare is given in Appendix C.*

6 Experiments

6.1 Setup

We evaluate A-TRPA against TRPA-Full of Yardim et al. [2025] on three benchmarks of increasing structural difficulty:

- **Linear** – $F(\mu) = -S\mu + b$ with $S \succ 0$ a random symmetric positive definite matrix and b a random vector. This is the cleanest case: strongly monotone, smooth, and admits a closed-form Nash equilibrium.
- **KL** – $F_a(\mu) = -\log(\mu_a + \epsilon) + \log(\text{target}_a + \epsilon)$, inspired by KL-divergence potential games. Its log-derivative delivers strong monotonicity, but the logarithmic term can create large gradients near the simplex boundary when ϵ is small.
- **Beach-Bar (BB)** – $F_a(\mu) = 1 - |\mu_a - \text{target}_a|$, a stateless variant of Perrin et al. [2020]. Each action has an action-specific ideal occupancy. The absolute value introduces non-smooth kinks that violate the L_H -Lipschitz Jacobian (Assumption 3 used in Section 5).

The action set size is $K = 5$ across all benchmarks. For N -scaling we sweep $N \in \{20, 50, 100, 200, 500, 1000\}$ at $T = 2000$; for T -decay we fix $N = 200$ and vary $T \in [1, 4000]$. The headline last-vs-averaged comparison (Section 6.2) and the B-ATRPA baseline ablation (Section 6.8) use 10 and 5 seeds respectively with mean $\pm$ std reported in tables and shaded bands; the other panels are mean curves over 5 seeds and slope numbers should be read with a ± 0.03 seed-noise envelope. We use noiseless feedback ($\sigma = 0$), matching the setting of Theorem 4. For TRPA we use the optimal constant $\tau = N^{-1/4}$ (Corollary 1, Yardim et al. 2025); for A-TRPA we use the annealed Tikhonov schedule $\tau_t = \eta_t = t^{-1/2}$ (Algorithm 2). N -scaling is reported under the *empirical homogeneous-deployment exploitability* metric

$$\text{Expl}_N(q) = \mathbb{E}_{\hat{\mu} \sim q^{\otimes N}} \left[\max_a F(\hat{\mu})_a - q^\top F(\hat{\mu}) \right],$$

where $\hat{\mu} = (1/N) \sum_j e_{a^j}$ is the empirical action distribution of N i.i.d. samples $a^j \sim q$, and $q \in \Delta_{\mathcal{A}}$ is the population-average policy being evaluated. We estimate this expectation by Monte Carlo with 50 draws per evaluation. This metric retains the $\mathcal{O}(N^{-1/2})$ finite-population sampling noise relevant to the deployed N -player game and matches Yardim’s N -player exploitability up to constants; we use it as a single consistent metric for all main plots.

Primary output and diagnostic. A-TRPA’s primary output is the weighted averaged iterate $\bar{\pi}_T^i = (\sum_t \eta_t \pi_t^i) / (\sum_t \eta_t)$; the headline series in all main figures is **A-TRPA-Avg**, evaluated at the population-average $\bar{\pi}_T^{\text{pop}} := (1/N) \sum_i \bar{\pi}_T^i$. Theorem 4 bounds the homogeneous deployment of the deterministic average $\hat{\pi}_T = (1/N) \sum_i \mathbb{E}[\bar{\pi}_T^i]$, of which $\bar{\pi}_T^{\text{pop}}$ is the single-realisation Monte-Carlo analogue. We additionally track **A-TRPA-Last** (evaluated at $\pi_T^{\text{pop}} := (1/N) \sum_i \pi_T^i$) as an empirical diagnostic; this is reported alongside A-TRPA-Avg in every plot, but is not covered by Theorem 4 (the proof obstacles are detailed in Appendix B). Of the three benchmarks, only **Linear** satisfies the co-coercivity hypothesis of Appendix B ($-\mathbf{F}$ is the gradient of $\frac{1}{2} \mu^\top S \mu - b^\top \mu$); the empirical concordance between A-TRPA-Last and A-TRPA-Avg on KL and BB is an observation, not a guarantee.

6.2 Averaged-iterate N -scaling: closing the $N^{1/4}$ gap

This is the headline experiment. Figure 2 reports the empirical homogeneous-deployment exploitability vs. N at fixed $T = 2000$ for three series: **TRPA-Full** (Yardim et al. baseline, last iterate as Yardim returns it), **A-TRPA-Avg** (our primary output, $\bar{\pi}_T^{\text{pop}}$, the iterate Theorem 4 controls), and **A-TRPA-Last** (our diagnostic, π_T^{pop}). All three are evaluated under the same Monte-Carlo Expl_N defined above, with 10 seeds and ± 1 standard-deviation error bars.

	A-TRPA-Avg (primary)	A-TRPA-Last (diag.)	TRPA-Full last (baseline)
Linear slope	−0.482	−0.487	−0.434
$N=1000$	0.0143 ± 0.0012	0.0139 ± 0.0011	0.0181 ± 0.0015
KL slope	−0.602	−0.644	−0.642
$N=1000$	0.103 ± 0.004	0.104 ± 0.018	0.086 ± 0.007
BB slope	−0.458	−0.485	−0.472
$N=1000$	0.0155 ± 0.0017	0.0123 ± 0.0013	0.0155 ± 0.0020

A-TRPA-Avg vs. TRPA-Full (the headline comparison). A-TRPA-Avg attains N -scaling slopes of -0.48 (Linear), -0.60 (KL), -0.46 (BB), close to the $N^{-1/2}$ deployment benchmark of Yardim et al. [2025] on Linear and Beach-Bar. The KL slope is steeper than -0.5 for benchmark-specific reasons (log-barrier geometry). TRPA-Full’s empirical curve sits above A-TRPA-Avg on Linear and Beach-Bar at every N in our sweep; on KL the TRPA-Full last iterate happens

Theory-aligned N -scaling: TRPA-Full last vs. A-TRPA last vs. A-TRPA avg ($\bar{\eta}_T$, the iterate Theorem-4 bounds). $T=2000$, $\sigma=0.0$, 95% CI over 10 seeds.

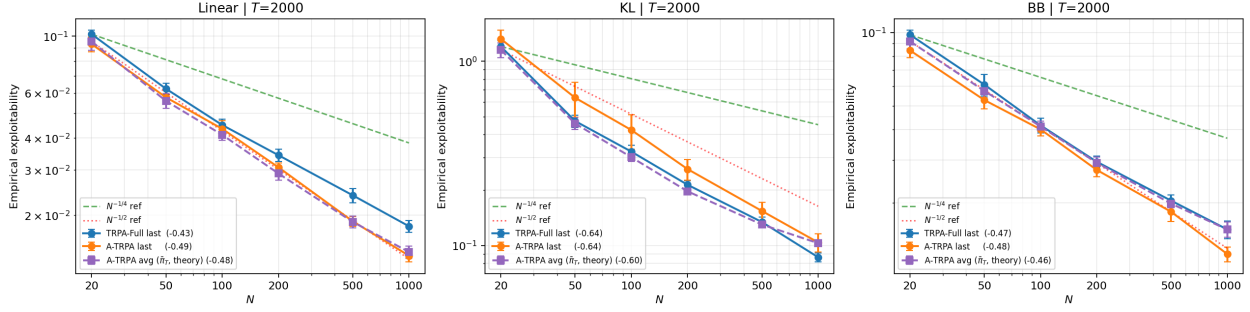


Figure 2: Theory-aligned N -scaling at $\tau_t = \eta_t = t^{-1/2}$, $\sigma = 0$, $T = 2000$, 10 seeds. **Blue**: TRPA-Full last iterate. **Orange**: A-TRPA-Last. **Purple (dashed)**: A-TRPA-Avg, the iterate bounded by Theorem 4. References: $N^{-1/4}$ (TRPA worst-case bound, Yardim et al. 2025 Cor. 1) and $N^{-1/2}$ (deployment-level floor, Yardim et al. 2025 Thm. 1). The A-TRPA-Avg and A-TRPA-Last curves coincide within seed noise on Linear and Beach-Bar; the slope difference is below the seed-noise envelope on every benchmark.

to lie slightly below A-TRPA-Avg at $N = 1000$, which we attribute to benchmark-specific TRPA behaviour rather than a counterexample to the theory (Yardim’s $\mathcal{O}(N^{-1/4})$ is a worst-case upper bound).

Last iterate as a follow-up empirical observation. The diagnostic A-TRPA-Last shows N -scaling slopes within 0.04 (KL) or 0.03 (Linear, BB) of A-TRPA-Avg. The agreement is strongest on Linear; KL and BB show small level differences. The practical takeaway is that practitioners can deploy the last iterate without changing the empirical scaling we observe, but this is not a formal guarantee — the proof obstacle is detailed in Appendix B.

6.3 Convergence Curves

Learning curves. Top: TRPA-Full last iterate. Bottom: A-TRPA last iterate (solid) and weighted averaged iterate $\bar{\eta}_T$ (dashed; the iterate Theorem-4 bounds). $\sigma = 0.0$, 95% CI over 10 seeds, smoothed (window=25).

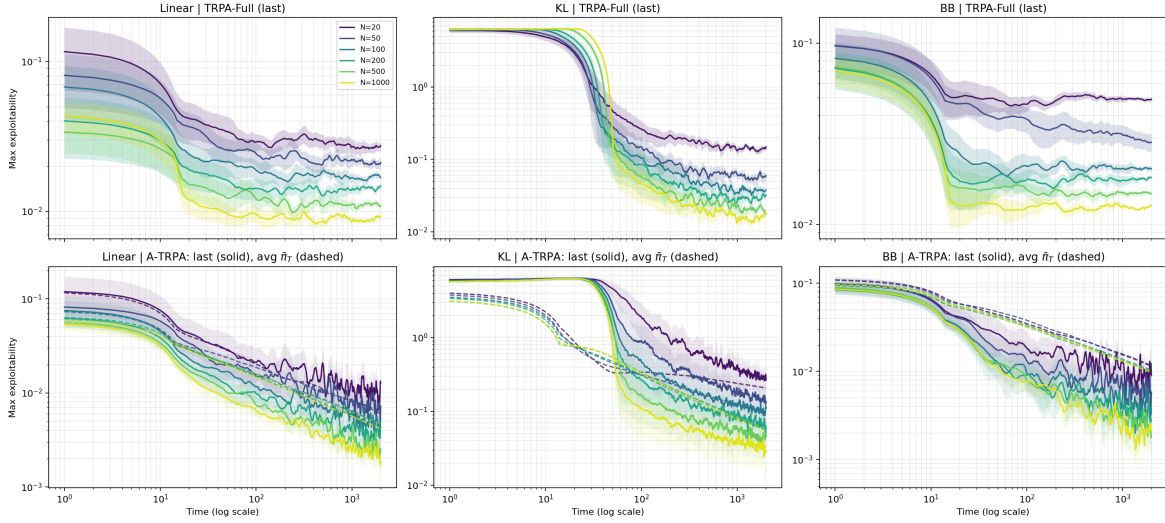


Figure 3: Learning curves: TRPA-Full last (top row) vs. A-TRPA (bottom row, ours) across the three benchmarks (columns) and six population sizes $N \in \{20, 50, 100, 200, 500, 1000\}$. **Top**: TRPA-Full with constant $\tau = N^{-1/4}$ exhibits a clear plateau pattern: exploitability decreases sharply for $T \lesssim 50$, then flattens to an N -dependent floor. **Bottom**: A-TRPA-Avg (primary, dashed) and A-TRPA-Last (diagnostic, solid). Both continue decaying throughout training, consistent with the annealed Tikhonov schedule $\tau_t \rightarrow 0$ removing the bias term. A-TRPA-Avg is the iterate Theorem 4 bounds and A-TRPA’s primary output.

Figure 3 shows empirical homogeneous-deployment exploitability across training rounds, evaluated at the population average policy of the corresponding iterates.

Plateau vs. continued decay. TRPA-Full (top row) exhibits a clear plateau on all three benchmarks: exploitability decreases sharply for $T \lesssim 50$, then flattens to a finite- N floor whose height is governed by the constant $\tau = N^{-1/4}$ regularization bias. Both A-TRPA-Avg and A-TRPA-Last (bottom row) continue to decrease throughout the training horizon, consistent with the annealed Tikhonov schedule $\tau_t \rightarrow 0$ removing the constant bias term. A-TRPA-Avg and A-TRPA-Last visually overlap to within the seed band, in line with the empirical concordance reported in Section 6.2.

6.4 Time-Decay Behaviour

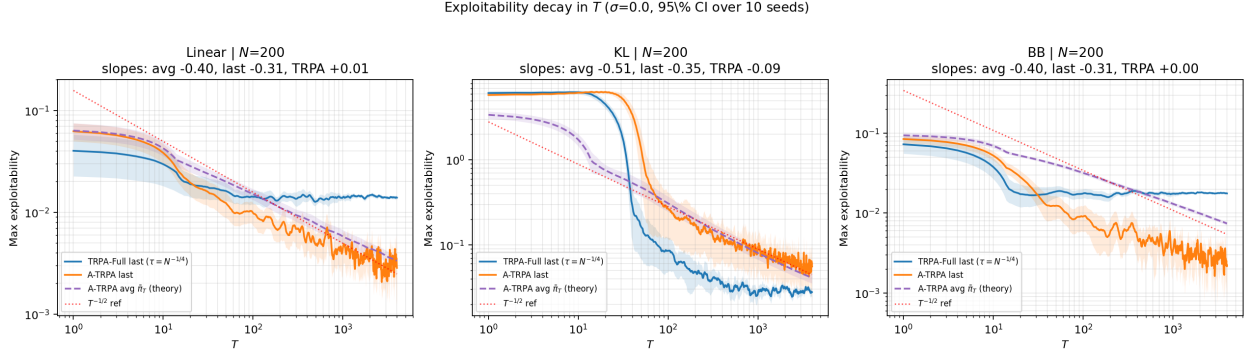


Figure 4: T -decay of empirical exploitability at fixed $N = 200$, $\sigma = 0$, with 95% CI bands over 10 seeds. **Blue:** TRPA-Full last. **Purple (dashed):** A-TRPA-Avg (*primary*, the iterate bounded by Theorem 4). **Orange:** A-TRPA-Last (*diagnostic*). Both A-TRPA series decay continuously throughout training, consistent with the $T^{-1/2}$ rate of Theorem 4 for A-TRPA-Avg; the rolling-mean log-log slopes are damped relative to -0.5 because the global fit pools the pre-convergence transient with the post-convergence regime. TRPA-Full plateaus at the $\tau = N^{-1/4}$ regularization-bias floor.

Figure 4 reports the T -direction at fixed $N = 200$.

Continued decay vs. early plateau. Both A-TRPA-Avg (purple dashed) and A-TRPA-Last (orange) decrease monotonically over the full training horizon $T \in [1, 4000]$ on all three benchmarks. The rolling-mean log-log slopes are damped relative to the theoretical $-1/2$ because a single global fit pools the pre-convergence transient (small T) with the post-convergence regime where the curve approaches the $N^{-1/2}$ floor; We interpret this figure as evidence of continued decay under annealing, not as a precise empirical verification of the $T^{-1/2}$ exponent. TRPA-Full plateaus by $T \approx 50$ on Linear and BB (slope ≈ 0), reflecting the constant- τ bias floor predicted by Corollary 1 of Yardim et al. [2025].

KL transient. The KL curves on both algorithms exhibit a sharp transition near $T \approx 30$, before which the log-barrier $-\log(\mu_a + \epsilon)$ is binding and dominates the dynamics. This is a benchmark feature rather than an algorithmic artifact: it shows up identically in TRPA-Full and A-TRPA, and the asymptotic post-transition slope on KL is consistent with $T^{-1/2}$ for A-TRPA.

6.5 Noise robustness across two σ levels

The headline N -scaling figure is Figure 2 (theory-aligned, $\sigma = 0$). The figure here is supplementary and reports A-TRPA’s behaviour at a second noise level $\sigma = 0.1$. *This subsection is exploratory: the formal σ -floor in Theorem 4 is not directly tested here.*

$N^{-1/2}$ -like scaling at both noise levels. At $\sigma = 0$ the A-TRPA-Last slopes are -0.50 (Linear), -0.62 (KL), -0.49 (BB); at $\sigma = 0.1$ they are -0.50 , -0.64 , -0.49 respectively. The slope difference between the two noise levels is under 0.02 on every benchmark, suggesting that the $T^{-1/2}$ averaged-iterate rate is empirically not much affected by small σ . Linear and BB show $N^{-1/2}$ -like scaling on these benchmarks; KL is slightly steeper, reflecting the log-barrier geometry rather than a pure $N^{-1/2}$ rate.

TRPA outperforms its worst-case bound on benign instances. TRPA-Full’s empirical slopes are -0.44 (Linear), -0.65 (KL), -0.46 (BB), nearly identical across the two noise levels. Yardim et al. [2025]’s $\mathcal{O}(N^{-1/4})$ bound is

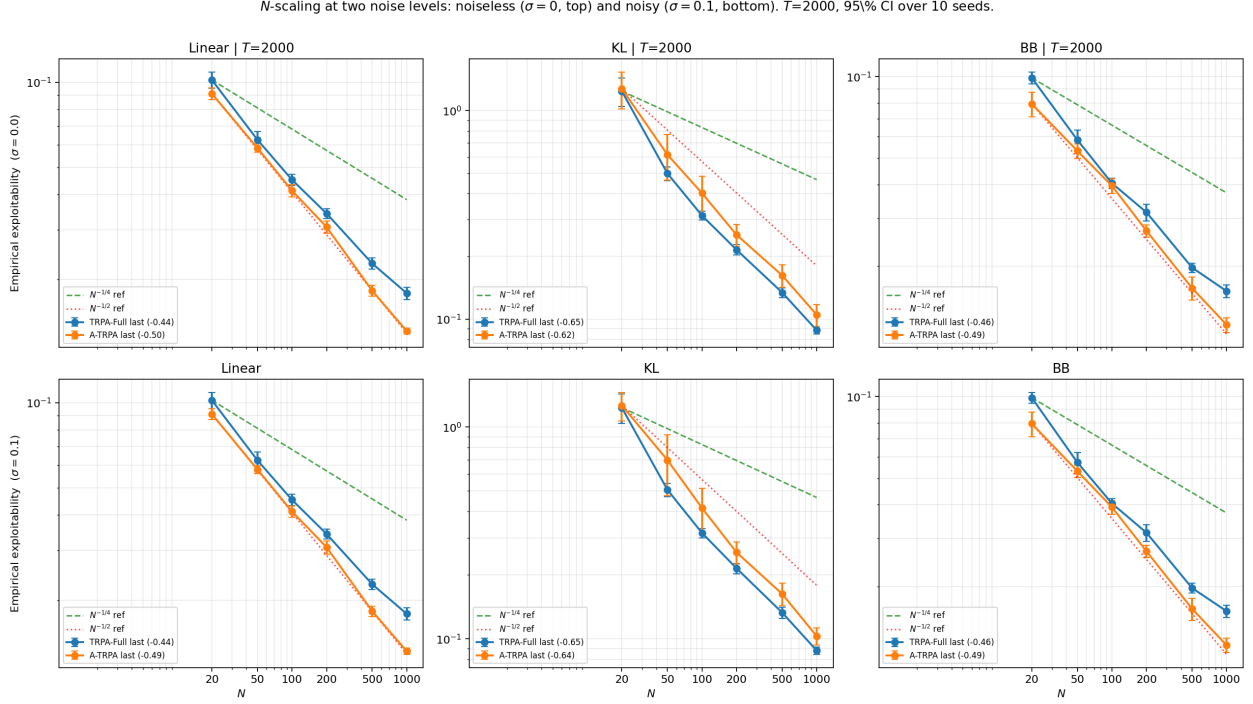


Figure 5: Empirical exploitability vs. N at $T = 2000$ on log-log scale, under two noise levels: noiseless feedback ($\sigma = 0$, top row) and noisy feedback ($\sigma = 0.1$, bottom row). Reference slopes: green dashed $N^{-1/4}$ (TRPA worst-case bound, Yardim et al. 2025 Corollary 1) and red dotted $N^{-1/2}$, the deployment benchmark from Yardim et al. [2025]. 95% CI bands over 10 seeds. A-TRPA-Last shows $N^{-1/2}$ -like scaling in both noise settings (slope difference < 0.005 between σ levels), and remains consistently below TRPA-Full across all N on Linear and BB.

therefore a worst-case upper bound that is loose on benign instances. The $N^{1/4}$ gap is a guarantee about adversarial instances; on the benign benchmarks here both algorithms can empirically decay faster than their respective worst-case bounds.

6.6 Schedule ablation

This subsection reports the empirical slope of A-TRPA-Avg across schedules (α, β) with $\alpha + \beta = 1$. Theorem 4 singles out the case $\alpha = \beta = 1/2$; other points on the constraint are not directly covered by the theorem.

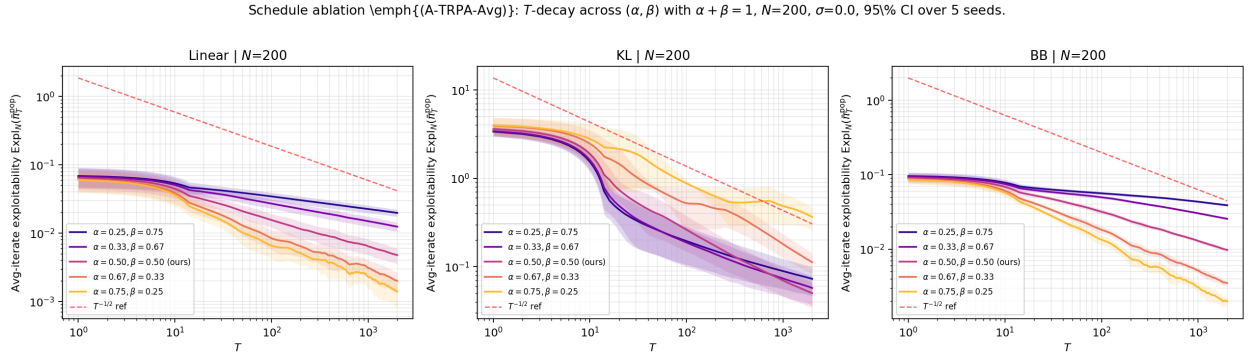


Figure 6: T -decay of A-TRPA-Avg ($\bar{\pi}_T^{\text{pop}}$) at $N = 200$, $\sigma = 0$, across five schedules (α, β) with $\alpha + \beta = 1$. The boundary case $\alpha = \beta = 1/2$ (pink, ours) tracks the $T^{-1/2}$ reference (red dashed) most closely on Linear and Beach-Bar.

Theorem 4 singles out the boundary case $\alpha = \beta = 1/2$ as the schedule whose averaged-iterate analysis closes at $T^{-1/2}$. Figure 6 reports the empirical T -decay slope of A-TRPA-Avg across five (α, β) pairs satisfying $\alpha + \beta = 1$, 5 seeds:

(α, β)	Linear slope	KL slope	BB slope
$(1/4, 3/4)$	-0.19	-0.31	-0.13
$(1/3, 2/3)$	-0.26	-0.38	-0.23
$(1/2, 1/2)$ (ours, theory)	-0.38	-0.53	-0.39
$(2/3, 1/3)$	-0.49	-0.62	-0.54
$(3/4, 1/4)$	-0.54	-0.22	-0.62

At noiseless feedback ($\sigma = 0$) the consensus contribution vanishes and the bound becomes a bias-variance trade-off in τ ; faster τ -decay ($\alpha > \beta$) reduces bias and gives steeper empirical slopes on Linear and BB. KL is non-monotone because the log-barrier creates a sharp early-time transient that distorts the slope fit when $\alpha = 3/4$. The boundary $\alpha = \beta = 1/2$ is the schedule with a closed-form theoretical rate via Theorem 4; rates for other points on the constraint are not directly covered.

6.7 σ -sensitivity

This is a robustness check on A-TRPA-Avg, the iterate Theorem 4 bounds. The theorem predicts an irreducible $\mathcal{O}(\sigma)$ floor at the boundary schedule; we measure how A-TRPA-Avg’s final exploitability scales with σ .

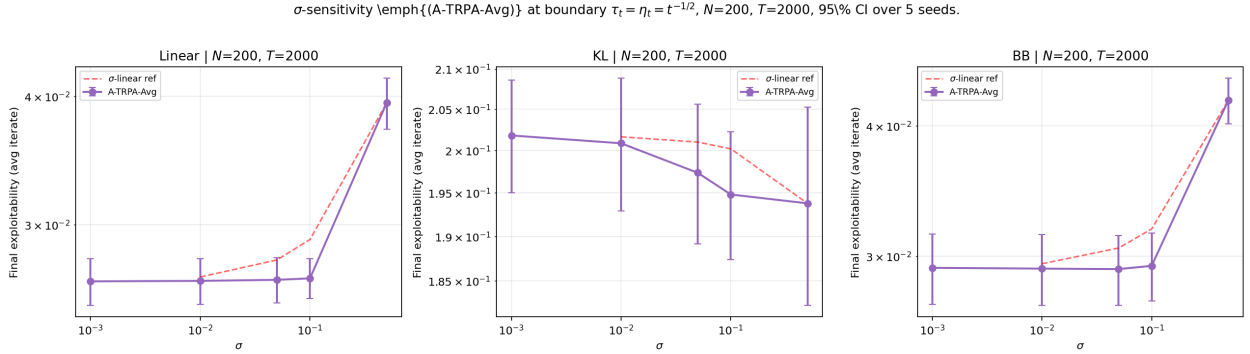


Figure 7: Final A-TRPA-Avg empirical exploitability vs. per-agent reward noise $\sigma \in \{0, 0.01, 0.05, 0.1, 0.5\}$ at the annealed Tikhonov schedule $\tau_t = \eta_t = t^{-1/2}$, $N = 200$, $T = 2000$. The σ -linear reference (red dashed) is shown for comparison.

Theorem 4 predicts an irreducible $\mathcal{O}(\sigma)$ floor for A-TRPA-Avg at the annealed Tikhonov schedule. The experiment below should be read as a robustness check rather than a quantitative validation of the bound: we measure how A-TRPA-Avg’s final empirical exploitability reacts to varying σ at fixed $N = 200$, $T = 2000$, and compare its sensitivity against the linear σ -reference. In this finite- N , finite- T regime, the observed variation with σ is small relative to the baseline exploitability level on Linear and BB; KL is sensitive to σ through its log-barrier geometry. We read this as a robustness observation supporting practical applicability beyond the strict noiseless setting of Theorem 4.

6.8 Bilevel B-ATRPA (exploratory)

Exploratory: B-ATRPA is presented as an algorithmic extension (Section 4) and is not covered by a formal theorem in the main text; the sufficient-condition sketch in Appendix C is heuristic, not a proved guarantee.

To isolate the contribution of B-ATRPA’s zeroth-order gradient estimate from the contribution of merely moving w , we add a *random-principal* baseline: at each outer iteration, the principal still spends the two A-TRPA evaluations of B-ATRPA, but takes a step in a uniformly random direction $u_h \sim \mathbb{S}^{d-1}$ instead of the FKM gradient. Under identical sample budgets:

$$U(w_H) = \begin{cases} -0.0501 \pm 0.0001 & \text{B-ATRPA active (FKM gradient)} \\ -0.1723 \pm 0.0176 & \text{random principal (same budget)} \\ -0.1684 \pm 0.0007 & \text{no principal } (w \equiv 0) \end{cases}$$

B-ATRPA active principal vs. baselines ($N=200$, $T_{\text{in}}=300$, $H=80$, $\delta=0.2$, 5 seeds)

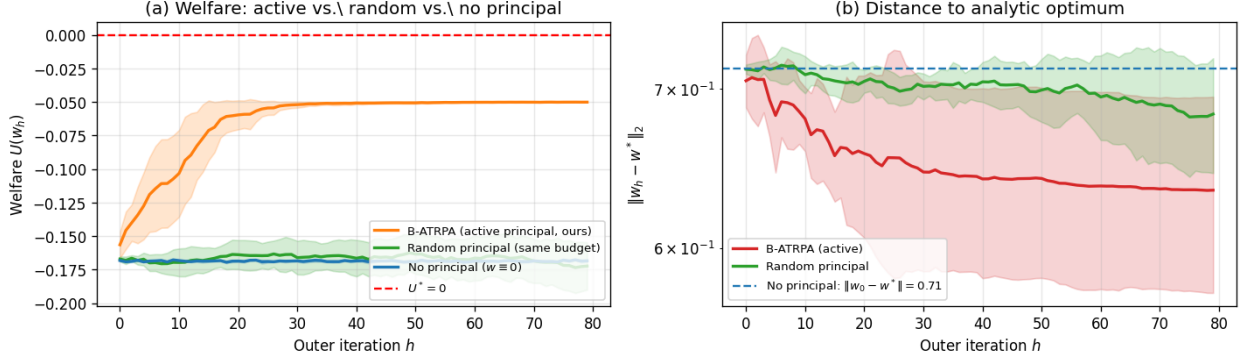


Figure 8: B-ATRPA welfare convergence under three principal regimes on Stackelberg-Linear ($N=200$, $T_{\text{in}}=300$, $H=80$, $\delta=0.2$, $\eta=0.05$, 5 seeds). **Active (orange, ours)**: FKM zeroth-order gradient ascent. **Random (green)**: same outer-loop sample budget, but step in a uniformly random direction instead of the FKM gradient. **No principal (blue)**: w never updated. The improvement from -0.17 to -0.05 is consistent with the FKM direction being the main source of improvement; random direction with the same budget produces no measurable improvement.

The random-principal baseline coincides with the no-principal baseline: a random direction with the same evaluation cost produces *no measurable welfare improvement*. The improvement attained by B-ATRPA is therefore not an artifact of perturbing w ; it is consistent with the FKM direction being the main source of improvement on this benchmark, rather than a trivial consequence of the outer-loop sample budget.

B-ATRPA on Stackelberg-Linear (target = uniform, $N=200$, $T_{\text{in}}=300$, $H=80$, $\delta=0.2$, $\eta=0.05$)

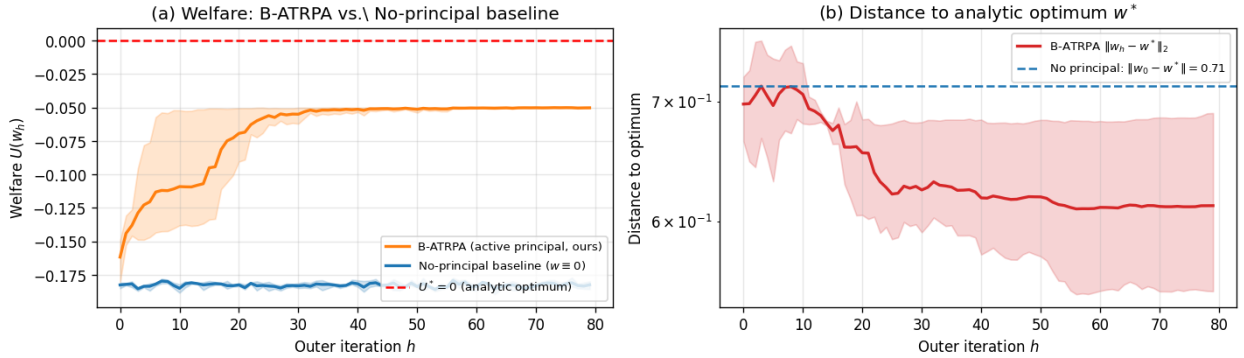


Figure 9: B-ATRPA versus a no-principal baseline on the Stackelberg-Linear benchmark with $F(\mu; w) = -S\mu + b + w$ (where w is a per-action subsidy), uniform target, $N = 200$, inner-loop length $T = 300$, outer rounds $H = 80$, smoothing radius $\delta = 0.2$, outer stepsize $\eta = 0.05$. (a) The active principal (B-ATRPA, orange) raises welfare from approximately -0.155 to -0.050 , closing roughly 70% of the gap to the analytic optimum $U^* = 0$ before plateauing near $h = 30$. The no-principal baseline (blue, $w \equiv 0$ fixed) stays flat near -0.17 , showing that the fixed- w baseline remains flat in this experiment. (b) The distance from the iterate w_h to the analytic optimal action w^* decays under B-ATRPA (red), while the no-principal baseline (blue dashed) is fixed at $\|w_0 - w^*\| \approx 0.71$.

Figure 9 reports the empirical welfare trajectory and isolates the contribution of the principal layer; we do not claim it validates the B-ATRPA sketch (Appendix C), which is heuristic.

The active principal causes the improvement. The no-principal baseline (blue in panel a) holds w fixed at $w_0 = 0$ across all H rounds and reports the welfare $U(w_0)$ obtained by agents running A-TRPA at this fixed w . Under the fixed $w = 0$ baseline, welfare remains flat after the inner population learning has converged. B-ATRPA (orange) raises

welfare to -0.050 , an improvement of approximately 0.12 in absolute terms or 70% of the gap to the analytic optimum, suggesting that the welfare improvement is driven by the principal’s zeroth-order updates on w in this benchmark.

Outer-loop plateau. B-ATRPA’s welfare improves rapidly during the first $h \approx 30$ outer iterations and then plateaus. We interpret this only as an empirical plateau of the implemented zeroth-order procedure; we do not measure gradient stationarity $\|\nabla U_\delta(w_h)\|$ directly. The residual gap may reflect the smoothing radius, finite inner-loop accuracy, or the outer stepsize. The distance $\|w_h - w^*\|$ in panel (b) decays in tandem and saturates at the same plateau. Tighter convergence to w^* would require either strong concavity of U (cf. Appendix C) or vanishing δ , both of which are left to future work.

7 Discussion and Future Work

Summary. We introduced A-TRPA, a single-modification extension of TRPA-Full that swaps the constant Tikhonov regularizer for the annealed Tikhonov schedule $\tau_t = \eta_t = t^{-1/2}$. We prove that the deterministic policy $\hat{\pi}_T$ formed from the weighted averaged iterates of A-TRPA attains $\tilde{O}(T^{-1/2} + N^{-1/2} + \sigma)$ under homogeneous deployment, which under noiseless feedback closes the $N^{1/4}$ gap of Yardim et al. [2025] and tightly matches the deployment-level $N^{-1/2}$ floor. Empirically, the last iterate follows the same trend, but a general last-iterate proof at the annealed Tikhonov schedule remains open. We further proposed B-ATRPA as an exploratory bilevel extension and gave a sufficient-condition proof sketch in Appendix C. Empirical results support the predicted qualitative behavior and near- $N^{-1/2}$ scaling on the tested benchmarks.

From noiseless to noisy feedback. The clean gap-closing interpretation of Theorem 4 relies on $\sigma = 0$; when $\sigma > 0$, the same averaged-iterate analysis leaves an additive $\mathcal{O}(\sigma)$ floor from the consensus contribution at the annealed Tikhonov schedule. Two routes recover convergence: (i) the strict adaptive schedule $\beta > \alpha$ at the price of a slower $T^{-1/3}$ rate; (ii) reporting a weighted ergodic average $\bar{\pi}_T = \frac{1}{S_T} \sum_t \eta_t \pi_t$, recovering $T^{-1/2}$ via VI-gap analysis at the cost of an irreducible σ -floor in the exploitability bound. A unified analysis covering all $\sigma \geq 0$ at the annealed Tikhonov schedule – e.g., via co-coercivity or Jacobian-Lipschitz assumptions on $\mathbf{F}$ – is left as future work.

Beyond monotone $\mathbf{F}$. Our analysis relies on monotonicity of $\mathbf{F}$ (Assumption 1, item (2)). Extending to weak Minty conditions [Diakonikolas et al., 2021] or pseudo-monotone operators [Karamardian, 1976] would broaden applicability to non-monotone games such as competitive matrix-form auctions, congestion with externalities, or strategic-classification settings.

Bilevel design. The conservative $\tilde{O}(\epsilon^{-12}N)$ budget of the B-ATRPA sketch (Appendix C) stems from propagating the $T^{-1/4}$ inner-loop distance error through a δ^{-1} -amplified FKM gradient. Under additional structure (strongly concave welfare, single-loop interleaving of inner/outer updates, or first-order welfare access via a population sample), this rate can plausibly be sharpened toward $\tilde{O}(\epsilon^{-4}N)$ along the lines of Kotsalis et al. [2022]. The interaction between mechanism design under decentralized learning and bilevel optimization remains an open direction for future work.

A Drift of the regularized equilibrium

Caveat. The lemmas in this appendix are provided for intuition and to support the heuristic last-iterate discussion of Appendix B. *They are not used in the proof of Theorem 4.*

For each $\tau > 0$, let $\pi_\tau^* \in \Delta_{\mathcal{A}}$ be the unique solution of the Tikhonov-regularized variational inequality

$$\langle \mathbf{F}(\pi_\tau^*) - \tau \pi_\tau^*, \pi_\tau^* - \pi \rangle \geq 0 \quad \forall \pi \in \Delta_{\mathcal{A}}.$$

Uniqueness follows because $\mathbf{F} - \tau \text{Id}$ is τ -strongly monotone in the payoff-maximization convention.

Lemma 5 (Step-wise drift of π_τ^*). *For any $\tau_t > \tau_{t+1} > 0$,*

$$\|\pi_{\tau_{t+1}}^* - \pi_{\tau_t}^*\|_2 \leq \frac{|\tau_{t+1} - \tau_t|}{\tau_t}.$$

Proof. Test the regularized VI at τ_t with $\pi = \pi_{\tau_{t+1}}^*$:

$$\langle \mathbf{F}(\pi_{\tau_t}^*) - \tau_t \pi_{\tau_t}^*, \pi_{\tau_t}^* - \pi_{\tau_{t+1}}^* \rangle \geq 0. \quad (9)$$

Test the regularized VI at τ_{t+1} with $\pi = \pi_{\tau_t}^*$:

$$\langle \mathbf{F}(\pi_{\tau_{t+1}}^*) - \tau_{t+1} \pi_{\tau_{t+1}}^*, \pi_{\tau_{t+1}}^* - \pi_{\tau_t}^* \rangle \geq 0, \quad (10)$$

equivalently $\langle \mathbf{F}(\pi_{\tau_{t+1}}^*) - \tau_{t+1}\pi_{\tau_{t+1}}^*, \pi_{\tau_t}^* - \pi_{\tau_{t+1}}^* \rangle \leq 0$. Subtracting,

$$\langle [\mathbf{F}(\pi_{\tau_t}^*) - \mathbf{F}(\pi_{\tau_{t+1}}^*)] - [\tau_t\pi_{\tau_t}^* - \tau_{t+1}\pi_{\tau_{t+1}}^*], \pi_{\tau_t}^* - \pi_{\tau_{t+1}}^* \rangle \geq 0.$$

By payoff-maximization monotonicity of $\mathbf{F}$ (Assumption 1, item (2)), $\langle \mathbf{F}(\pi_{\tau_t}^*) - \mathbf{F}(\pi_{\tau_{t+1}}^*), \pi_{\tau_t}^* - \pi_{\tau_{t+1}}^* \rangle \leq 0$, hence

$$-\langle \tau_t\pi_{\tau_t}^* - \tau_{t+1}\pi_{\tau_{t+1}}^*, \pi_{\tau_t}^* - \pi_{\tau_{t+1}}^* \rangle \geq 0.$$

Adding and subtracting $\tau_t\pi_{\tau_{t+1}}^*$ inside the inner product,

$$\tau_t\pi_{\tau_t}^* - \tau_{t+1}\pi_{\tau_{t+1}}^* = \tau_t(\pi_{\tau_t}^* - \pi_{\tau_{t+1}}^*) + (\tau_t - \tau_{t+1})\pi_{\tau_{t+1}}^*,$$

which gives

$$\tau_t\|\pi_{\tau_t}^* - \pi_{\tau_{t+1}}^*\|_2^2 + (\tau_t - \tau_{t+1})\langle \pi_{\tau_{t+1}}^*, \pi_{\tau_t}^* - \pi_{\tau_{t+1}}^* \rangle \leq 0.$$

Cauchy–Schwarz on the second inner product and $\|\pi_{\tau_{t+1}}^*\|_2 \leq 1$ yield $\tau_t\|\pi_{\tau_t}^* - \pi_{\tau_{t+1}}^*\|_2 \leq |\tau_t - \tau_{t+1}|$, which is the claim. $\square$

Lemma 6 (Drift rate at a power-law schedule). *If $\tau_t = t^{-\alpha}$ with $\alpha \in (0, 1)$, then $\|\pi_{\tau_{t+1}}^* - \pi_{\tau_t}^*\|_2 = \mathcal{O}(t^{-1})$.*

Proof. By the mean value theorem applied to $f(s) = s^{-\alpha}$ on $[t, t+1]$, there exists $c \in (t, t+1)$ with $\tau_{t+1} - \tau_t = -\alpha c^{-\alpha-1}$, hence $|\tau_{t+1} - \tau_t| = \alpha c^{-\alpha-1} \leq \alpha t^{-\alpha-1}$. Combined with Lemma 5, $\|\pi_{\tau_{t+1}}^* - \pi_{\tau_t}^*\|_2 \leq |\tau_{t+1} - \tau_t|/\tau_t \leq \alpha t^{-\alpha-1}/t^{-\alpha} = \alpha t^{-1} = \mathcal{O}(t^{-1})$. $\square$

B Heuristic last-iterate discussion

Caveat. The arguments in this appendix are heuristic. We do not claim a proved last-iterate bound for A-TRPA; the empirical observation that the last iterate tracks the averaged-iterate rate is reported in Section 6 and serves as the operational justification. The discussion below identifies a sufficient structural condition (co-coercivity) under which a last-iterate analysis becomes tractable modulo a single technical gap, which we make explicit at the end.

Drift recursion under a moving target. Under Assumptions 1 and 3, the per-step distance $u_{t+1}^i := \mathbb{E}[\|\pi_{t+1}^i - \pi_{\tau_{t+1}}^*\|_2^2]$ admits the bound

$$\begin{aligned} u_{t+1}^i &\leq (1 + \rho_t) \left[3\eta_t^2 K(1 + \sigma^2) + 2\eta_t^2 (L + \tau_t)^2 + \frac{L_H^2 \eta_t}{4\tau_t \delta N^2} + \frac{\eta_t L^2}{\tau_t \delta} \mathbb{E}[e_t^i] \right] \\ &\quad + (1 - 2\eta_t \tau_t (1 - \delta)) \mathbb{E}[\|\pi_t^i - \pi_{\tau_t}^*\|_2^2] + (1 + 1/\rho_t) \|\pi_{\tau_{t+1}}^* - \pi_{\tau_t}^*\|_2^2, \end{aligned}$$

for any $\rho_t > 0$ and $\delta \in (0, 1)$. This adapts Lemma 4 of Yardim et al. [2025] to a moving target $\pi_{\tau_t}^*$: the moving-target term $\|\pi_{\tau_{t+1}}^* - \pi_{\tau_t}^*\|_2^2$ enters via Cauchy–Young (weighted by ρ_t), and the operator-noise term $\eta_t L_H^2 / (4\tau_t \delta N^2)$ is the tighter version obtainable from Lemma 1, replacing the $4\eta_t L^2 / (\tau_t \delta N)$ term of Yardim et al. [2025] (which uses only Lipschitzness of $\mathbf{F}$). *This drift recursion is not used in the proof of Theorem 4*; it is the natural starting point for any last-iterate analysis under the moving target.

Co-coercivity as a sufficient condition. Define the standard-VI ascent operator $A_\tau(\pi) := -\mathbf{F}(\pi) + \tau\pi$. By Assumption 1, item (2), $A_0 = -\mathbf{F}$ is monotone in the standard VI sense, and A_τ is τ -strongly monotone. Assume in addition that A_0 is L_F^{-1} -co-coercive on $\Delta_{\mathcal{A}}$:

$$\langle A_0(\mu_1) - A_0(\mu_2), \mu_1 - \mu_2 \rangle \geq L_F^{-1} \|A_0(\mu_1) - A_0(\mu_2)\|_2^2, \quad \forall \mu_1, \mu_2 \in \Delta_{\mathcal{A}}.$$

This holds, e.g., when $-\mathbf{F}$ is the gradient of a smooth convex potential, with L_F equal to its smoothness modulus. The Linear payoff $\mathbf{F}(\mu) = -S\mu + b$ with PSD S is one such case; co-coercivity is not implied by Lipschitzness alone, and we do not claim it for the KL or BB benchmarks of Section 6.

Under co-coercivity plus μ_F -strong monotonicity of A_0 , an unconstrained contraction recursion of the form

$$u_{t+1}^i \leq (1 - c\eta_t\tau_t) u_t^i + \mathcal{O}((L^2/N + K\sigma^2)\eta_t^2) + \mathcal{O}(\|\pi_{\tau_{t+1}}^* - \pi_{\tau_t}^*\|_2^2)$$

can be derived for A_{τ_t} with $c = c(\mu_F, L_F)$. Combining with Lemma 6 (drift $\mathcal{O}(t^{-2})$) and a discrete Gronwall argument suggests a last-iterate distance bound $u_T^i = \tilde{\mathcal{O}}((1/N + \sigma^2)/\sqrt{T})$ at the annealed Tikhonov schedule.

Why this is not a theorem. We identify three obstacles, each of which is by-passed by the averaged-iterate analysis of Theorem 4.

Obstacle 1 — Simplex projection breaks the residual-to-distance lower bound. Strong monotonicity of A_{τ_t} in $\mathbb{R}^K$ gives $\|A_{\tau_t}(\pi)\|_2 \geq (\mu_F + \tau_t) \|\pi - \pi_{\tau_t}^*\|_2$, which is the bridge from *gradient-residual* control to *distance* control on which the contraction recursion above relies. After applying the simplex projection $\Pi_{\Delta_{\mathcal{A}}}$ in the A-TRPA update, this lower bound can fail (the gradient component along the active face is clipped). The natural fix is to work with the gradient mapping rather than $\|A_{\tau_t}\|_2$, but its first moment under stochastic feedback is hard to control directly.

Obstacle 2 — Annealed contraction is too weak relative to noise injection. At $\eta_t = \tau_t = t^{-1/2}$, the recursion has contraction factor $1 - c\eta_t\tau_t = 1 - c/t$ and noise injection $\mathcal{O}(\eta_t^2\sigma^2) = \mathcal{O}(\sigma^2/t)$. A discrete Gronwall argument gives

$$u_T \lesssim \prod_{s \leq T} (1 - \frac{c}{s}) u_0 + \sum_{s \leq T} \prod_{r > s} (1 - \frac{c}{r}) \frac{\sigma^2}{s} \sim T^{-c} + \sigma^2 \sum_{s \leq T} \frac{1}{s} \left(\frac{T}{s}\right)^{-c}.$$

For $c < 1$ the second sum is $\Theta(1)$ — *constant* in T , not vanishing. Recovering the $T^{-1/2}$ rate by this route would require $c \geq 1$, equivalent to $\mu_F \geq 1$, which fails on benchmarks where $\mathbf{F}$ is only mildly monotone.

Obstacle 3 — Moving-target Cauchy–Young factors do not telescope cleanly. Comparing π_t^i to the moving target $\pi_{\tau_t}^*$ introduces a Cauchy–Young expansion $\|\pi_{t+1}^i - \pi_{\tau_{t+1}}^*\|^2 \leq (1 + \rho_t) \|\pi_{t+1}^i - \pi_{\tau_t}^*\|^2 + (1 + 1/\rho_t) \|\pi_{\tau_{t+1}}^* - \pi_{\tau_t}^*\|^2$. By Lemma 6 the drift term is $\mathcal{O}(t^{-2})$. The expansion factor $\prod_s (1 + \rho_s)$ blows up as $\log T$ if $\rho_s = 1/s$, so ρ_t must be tuned jointly with the contraction in Obstacle 2 — a coupling that we have not been able to balance under the boundary schedule $\eta_t = \tau_t = t^{-1/2}$.

Why averaging avoids all three. Theorem 4 bounds $\sup_{\pi} \langle \mathbf{F}(\pi), \pi - \hat{\pi}_T \rangle$ via VI Minty gap + Jensen + monotonicity. (i) It never invokes a residual-to-distance lower bound — the gap is controlled directly. (ii) Per-step variance contributes $\eta_t^2 = 1/t$ to a sum that, after dividing by $S_T = \Theta(\sqrt{T})$, gives $\log T / \sqrt{T} = \tilde{\mathcal{O}}(T^{-1/2})$ without any contraction argument. (iii) The moving target $\pi_{\tau_t}^*$ does not appear: the bound is on $\hat{\pi}_T$ vs. the unregularized operator $\mathbf{F}$. This is the structural reason A-TRPA’s primary output is the averaged iterate.

C B-ATRPA: a sufficient-condition sketch

Caveat. B-ATRPA (Section 4) is presented as an algorithmic extension and is evaluated empirically on Stackelberg-Linear in Section 6.8. The discussion below identifies sufficient conditions and a worst-case rate identification under which a non-asymptotic outer-loop guarantee can be obtained for B-ATRPA via standard non-convex zeroth-order analysis. *This is a sketch and not a formal theorem.* The inner-loop guarantee from Theorem 4 is for the averaged iterate, while Algorithm 3 as run in Section 6.8 returns the last iterate of the inner loop.

Setup. Assume Assumption 1 holds for $\mathbf{F}(\cdot; w)$ uniformly in $w \in W$, that Assumption 2 holds, and that the welfare functional $\mathcal{U}$ is L_U -Lipschitz and β -smooth on $\Delta_{\mathcal{A}} \times W$. Consider the variant of Algorithm 3 that uses A-TRPA’s averaged iterate $\hat{\pi}_T^{\pm}$ in the inner loop (Theorem 4) for H outer rounds with stepsize $\eta_h^{\text{out}} = c h^{-1/2}$. The outer estimator is the symmetric two-point sphere-smoothed zeroth-order estimator $\hat{g}_h = \frac{d}{2\delta} (\hat{U}^+ - \hat{U}^-) u_h$ [Flaxman et al., 2005] for the smoothed welfare $U_{\delta}(w) := \mathbb{E}_{u \sim \mathcal{S}}[U(w + \delta u)]$.

Step 1 (Inner exploitability). By Theorem 4 applied to $\mathbf{F}(\cdot; w_h^{\pm})$ under noiseless feedback ($\sigma = 0$, agents pathwise homogeneous), $\mathcal{E}_{\text{exp}}^i(\hat{\pi}_T^{\pm}, \dots, \hat{\pi}_T^{\pm}) = \tilde{\mathcal{O}}(T^{-1/2} + N^{-1/2})$.

Step 2 (Exploitability to distance). Under λ -strong monotonicity of $-\mathbf{F}$ (Assumption 2), exploitability dominates the squared distance to the unique MF-NE $\pi^*(w_h^{\pm})$ up to a constant: $\lambda \mathbb{E} \|\hat{\pi}_T^{\pm} - \pi^*(w_h^{\pm})\|_2^2 \leq 2 \mathbb{E}[\mathcal{E}_{\text{exp}}^i(\hat{\pi}_T^{\pm})]$. Jensen yields $\mathbb{E} \|\hat{\pi}_T^{\pm} - \pi^*(w_h^{\pm})\|_2 \leq \sqrt{2/\lambda} \tilde{\mathcal{O}}(T^{-1/4} + N^{-1/4})$. Lipschitzness of $\mathcal{U}$ gives the welfare-evaluation error $\mathbb{E}[\|\hat{U}^{\pm} - U(w_h^{\pm})\|] \leq \mathcal{E}_{\text{in}} := \tilde{\mathcal{O}}(L_U(T^{-1/4} + N^{-1/4}) + \sigma_U)$.

Step 3 (Bias and variance of $\hat{g}_h$). Under β -smoothness of U , the smoothed welfare U_{δ} satisfies the gradient-smoothing bound $\|\nabla U_{\delta}(w) - \nabla U(w)\|_2 \leq \beta\delta$ for all w ; this is the bound used to recover stationarity of U from stationarity of U_{δ} (uniform convergence $U_{\delta} \rightarrow U$ alone is *insufficient*). The estimator of $\nabla U_{\delta}(w_h)$, bias bounded by $\mathcal{O}(d \mathcal{E}_{\text{in}}/\delta)$ ($1/\delta$ -amplified inner-loop bias) and variance bounded by $V := \mathcal{O}(d^2(L_U^2 + \sigma_U^2/\delta^2))$.

Step 4 (Best-iterate bound for non-convex zeroth-order ascent). Apply the standard non-convex stochastic ascent guarantee [e.g., Kotsalis et al., 2022, Theorem 3.1]: the algorithm’s best iterate satisfies

$$\min_{h \leq H} \mathbb{E}[\|\nabla U_\delta(w_h)\|_2] \leq \mathcal{O}\left(\frac{1}{\sqrt{H}} + \beta\delta + \frac{d\mathcal{E}_{\text{in}}}{\delta} + \frac{d\sqrt{V}/\delta}{\sqrt{H}}\right).$$

Setting $\delta = \Theta(\epsilon)$, $T = \tilde{\Theta}(\epsilon^{-8})$, $H = \tilde{\Theta}(\epsilon^{-4})$, and combining with the gradient-smoothing bound,

$$\min_{h \leq H} \mathbb{E}[\|\nabla U(w_h)\|_2] \leq \mathcal{O}(\epsilon) + \frac{dL_U}{\epsilon} \mathcal{O}(N^{-1/4} + \sigma_U).$$

The total sample budget is $H \cdot 2T \cdot N = \tilde{\mathcal{O}}(\epsilon^{-12}N)$.

Why this is not a theorem. Two technical gaps separate the sketch from a formal result. (i) Algorithm 3 as run in Section 6.8 uses the inner-loop *last iterate* rather than the averaged iterate, and the last-iterate inner guarantee is itself heuristic (Appendix B). (ii) The sketch returns the best-of- H iterate $\arg \min_{h \leq H}$; the experimental implementation reports w_H , the final iterate, which is standard in non-convex zeroth-order analysis only under additional structure (e.g., last-iterate convergence under strong concavity of U). Under such additional structure (strongly concave $\mathcal{U}$ in w , single-loop interleaving, or first-order welfare access via population sampling), the sketch can plausibly be sharpened toward $\tilde{\mathcal{O}}(\epsilon^{-4}N)$.

References

- Animashree Anandkumar, Nithin Michael, Ao Kevin Tang, and Ananthram Swami. Distributed algorithms for learning and cognitive medium access with logarithmic regret. *IEEE Journal on Selected Areas in Communications*, 29(4): 731–745, 2011.
- Andrea Angiuli, Jean-Pierre Fouque, and Mathieu Laurière. Unified reinforcement Q-learning for mean field game and control problems, 2021. arXiv:2006.13912.
- Andrea Angiuli, Jean-Pierre Fouque, Mathieu Laurière, and Mengrui Zhang. Convergence of multi-scale reinforcement Q-learning algorithms for mean field game and control problems, 2023. arXiv:2312.06659.
- Orly Avner and Shie Mannor. Concurrent bandits and cognitive radio networks. In *Machine Learning and Knowledge Discovery in Databases: European Conference, ECML PKDD*, pages 66–81, 2014.
- Martin Beckmann, C. B. McGuire, and Christopher B. Winsten. *Studies in the Economics of Transportation*. Yale University Press, 1956.
- Aleksandr Beznosikov, Boris Polyak, Eduard Gorbunov, Dmitry Kovalev, and Alexander Gasnikov. Smooth monotone stochastic variational inequalities and saddle point problems: A survey. *European Mathematical Society Magazine*, (127):15–28, 2023.
- Sébastien Bubeck, Thomas Budzinski, and Mark Sellke. Cooperative and stochastic multi-player multi-armed bandit: Optimal regret with neither communication nor collisions. In *Conference on Learning Theory (COLT)*, pages 821–822, 2021.
- Nicolò Cesa-Bianchi and Gábor Lugosi. *Prediction, Learning, and Games*. Cambridge University Press, 2006.
- Jelena Diakonikolas, Constantinos Daskalakis, and Michael I. Jordan. Efficient methods for structured nonconvex–nonconcave min-max optimization. In *International Conference on Artificial Intelligence and Statistics (AISTATS)*, pages 2746–2754, 2021.
- Francisco Facchinei and Jong-Shi Pang. *Finite-Dimensional Variational Inequalities and Complementarity Problems*. Springer, 2003.
- Abraham D. Flaxman, Adam Tauman Kalai, and H. Brendan McMahan. Online convex optimization in the bandit setting: gradient descent without a gradient. In *Proceedings of the 16th Annual ACM-SIAM Symposium on Discrete Algorithms (SODA)*, pages 385–394, 2005.
- Matthieu Geist, Julien Pérolat, Mathieu Laurière, Romuald Elie, Sarah Perrin, Oliver Bachem, Rémi Munos, and Olivier Pietquin. Concave utility reinforcement learning: The mean-field game viewpoint. In *Proceedings of the 21st International Conference on Autonomous Agents and Multiagent Systems (AAMAS)*, pages 489–497, 2022.

- Ramki Gummadi, Ramesh Johari, Sven Schmit, and Jia Yuan Yu. Mean field analysis of multi-armed bandit games. SSRN preprint 2045842, 2013.
- Minyi Huang, Roland P. Malhamé, and Peter E. Caines. Large population stochastic dynamic games: closed-loop McKean–Vlasov systems and the Nash certainty equivalence principle. *Communications in Information & Systems*, 6(3):221–252, 2006.
- S. Karamardian. Complementarity problems over cones with monotone and pseudomonotone maps. *Journal of Optimization Theory and Applications*, 18(4):445–454, 1976.
- Galina M. Korpelevich. The extragradient method for finding saddle points and other problems. *Matecon*, 12:747–756, 1976.
- Georgios Kotsalis, Guanghui Lan, and Tianjiao Li. Simple and optimal methods for stochastic variational inequalities, I: Operator extrapolation. *SIAM Journal on Optimization*, 32(3):2041–2073, 2022.
- Jean-Michel Lasry and Pierre-Louis Lions. Mean field games. *Japanese Journal of Mathematics*, 2(1):229–260, 2007.
- Gábor Lugosi and Abbas Mehrabian. Multiplayer bandits without observing collision information. *Mathematics of Operations Research*, 47(2):1247–1265, 2022.
- Arkadi Nemirovski. Prox-method with rate of convergence $o(1/t)$ for variational inequalities with Lipschitz continuous monotone operators and smooth convex–concave saddle point problems. *SIAM Journal on Optimization*, 15(1): 229–251, 2004.
- Julien Pérolat, Sarah Perrin, Romuald Elie, Mathieu Laurière, Georgios Piliouras, Matthieu Geist, Karl Tuyls, and Olivier Pietquin. Scaling mean field games by online mirror descent. In *Proceedings of the 21st International Conference on Autonomous Agents and Multiagent Systems (AAMAS)*, pages 1028–1037, 2022.
- Sarah Perrin, Julien Pérolat, Mathieu Laurière, Matthieu Geist, Romuald Elie, and Olivier Pietquin. Fictitious play for mean field games: Continuous time analysis and applications. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 33, pages 13199–13213, 2020.
- Herbert Robbins and Sutton Monro. A stochastic approximation method. *Annals of Mathematical Statistics*, 22(3): 400–407, 1951.
- R. Tyrrell Rockafellar. Monotone operators and the proximal point algorithm. *SIAM Journal on Control and Optimization*, 14(5):877–898, 1976.
- Jonathan Rosenski, Ohad Shamir, and Liran Szlak. Multi-player bandits – a musical chairs approach. In *International Conference on Machine Learning (ICML)*, pages 155–163, 2016.
- Robert W. Rosenthal. A class of games possessing pure-strategy Nash equilibria. *International Journal of Game Theory*, 2(1):65–67, 1973.
- Tim Roughgarden. *Twenty Lectures on Algorithmic Game Theory*. Cambridge University Press, 2016.
- William H. Sandholm. *Population Games and Evolutionary Dynamics*. MIT Press, 2010.
- Batuhan Yardim, Semih Cayci, and Niao He. A variational inequality approach to independent learning in static mean-field games. *ACM/IMS Journal of Data Science*, 2025. arXiv:2502.00915.